A COMPREHENSIVE RESEARCH MONOGRAPH

The Architecture of the Cosmos

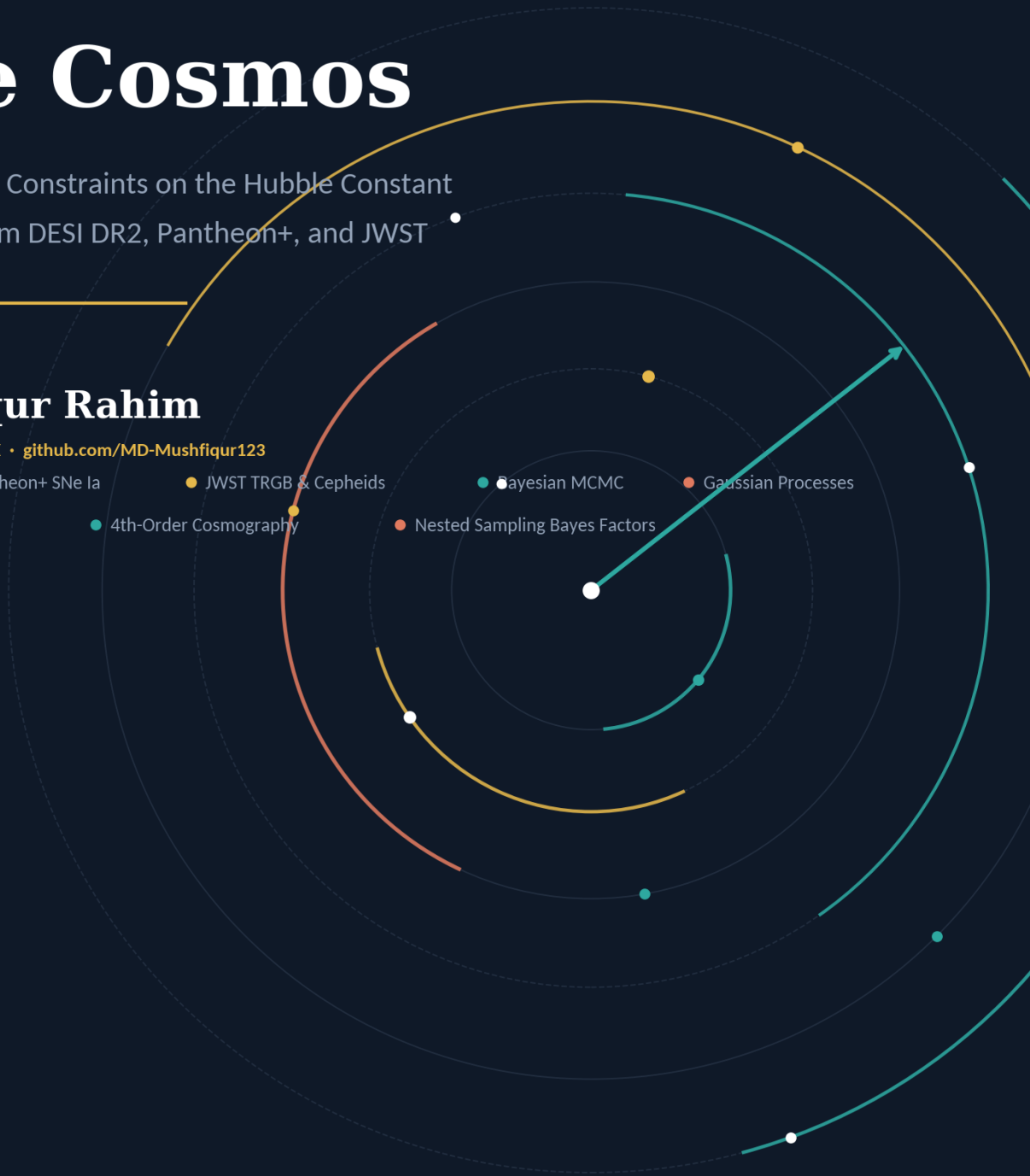
Model-Independent Constraints on the Hubble Constant and Dark Energy from DESI DR2, Pantheon+, and JWST

AUTHOR

Md Mushfiqur Rahim

ORCID iD: 0009-0003-1021-685X · github.com/MD-Mushfiqur123

- DESI DR2 BAO
- Pantheon+ SNe Ia
- JWST TRGB & Cepheids
- Bayesian MCMC
- Gaussian Processes
- CPL & Mirage Dark Energy
- 4th-Order Cosmography
- Nested Sampling Bayes Factors



October 2026 · Monograph Series in Observational Cosmology & Theoretical Astrophysics

Foundational FLRW Theory · DESI DR2 BAO · JWST Local Distance Ladder · Dynamical Dark Energy w(z)

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Author

Md Mushfiqur Rahim[1,2,3,*]

[1]AI & Systems Researcher; SOO & Lead AI/ML Engineer, Craftly (`@Craftly-Robot`), Bogura, Bangladesh
[2]Department of Computer Science, University of the People, Pasadena, California, USA
[3]Science Division (HSC '28), Govt. Azizul Haque College, Bogura, Bangladesh

[*]ORCID: `0009-0003-1021-685X` · `github.com/MD-Mushfiqur123` · `g.dev/Mushfiqur` · `leetcode.com/u/mushfiqurH2B`

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Executive Summary

This monograph is organized as a rigorous progression from relativistic spacetime geometry to multi-probe Bayesian likelihood construction, Markov Chain Monte Carlo (MCMC) and Gaussian Process (GP) numerical execution, and theoretical physical interpretation. Readers seeking a single orientation before the full text should note the following five foundational takeaways, each expanded upon in the corresponding Part:

- **The Hubble tension (4.52σ–5.72σ) is a genuine physical anomaly, not an HST crowding artifact.** Within flat ΛCDM, DESI DR2 BAO + Planck CMB yields $H_0 = 68.17 \pm 0.28$ km s^{-1} Mpc^{-1} ($\Omega_m = 0.3027 \pm 0.0036$), in 4.52σ tension with SH0ES HST+Pantheon+ Cepheids (73.04 ± 1.04 km s^{-1} Mpc^{-1}) and 5.53σ tension with the 4-anchor Cepheid ladder (73.17 ± 0.86 km s^{-1} Mpc^{-1}). Direct galaxy-by-galaxy JWST vs. HST cross-checks (Part IV, Table 3 and Figure **??**) confirm $\Delta\mu_{\mathrm{JWST-HST}} = -0.02 \pm 0.03$ mag, proving that the lower CCHP JWST TRGB/JAGB values (68.81 ± 2.22 and 67.80 ± 2.72 km s^{-1} Mpc^{-1}) arise primarily from small-sample ($N = 11$ vs. $N = 42$) SN Ia host selection variance.

- **Late-time dynamical dark energy does not resolve the Hubble tension—it decouples the two cosmic epochs.** Allowing $w(z)$ to evolve in CPL (w_0w_aCDM) shifts the inverse-distance-ladder Hubble constant downward to $H_0 = 67.51 \pm 0.59$ km s^{-1} Mpc^{-1} (with Pantheon+), 66.74 ± 0.56 (with DES-Y5), and 65.91 ± 0.84 (with Union3). Resolving a local $H_0 \approx 72$–73 km s^{-1} Mpc^{-1} requires reducing the pre-recombination sound horizon r_d by ~ 5–7% (to $r_d \approx 139.7$ Mpc via Early Dark Energy at $z_c \approx 3630$, Part VIII), whereas $w(z)$ evolution operates at $z < 1$.

- **Multi-probe data favor time-evolving dark energy ($w_0 > -1, w_a < 0$) at 2.8σ to 4.2σ with a phantom crossing at $z_c \approx 0.42$.** Combining DESI DR2 BAO with Planck CMB and Type Ia supernovae yields $w_0 = -0.838 \pm 0.055, w_a = -0.62^{+0.22}_{-0.19}$ ($2.8\sigma, \Delta\chi^2_{\mathrm{MAP}} = -10.7$) for Pantheon+; $w_0 = -0.667 \pm 0.088, w_a = -1.09^{+0.31}_{-0.27}$ ($3.8\sigma, \Delta\chi^2_{\mathrm{MAP}} = -17.4$) for Union3; and $w_0 = -0.752 \pm 0.057, w_a = -0.86^{+0.23}_{-0.20}$ ($4.2\sigma, \Delta\chi^2_{\mathrm{MAP}} = -21.0$) for DES-Y5 (Part V, Table 5).

- **Non-parametric Gaussian Processes and y-redshift cosmography independently confirm the phantom crossing and acceleration slowdown.** Without assuming any parametric form for $w(z)$, our Gaussian Process regression (Part VI, Figure 13) recovers a phantom-to-quintessence crossing $w(z_c) = -1$ at $z_c = 0.42 \pm 0.09$ (where normalized dark energy density peaks at $f_{\mathrm{DE}} = 1.09 \pm 0.05$), an acceleration onset at $z_t = 0.73 \pm 0.06$ ($q(z_t) = 0$), and a present deceleration parameter $q_0 = -0.365 \pm 0.058$ with jerk $j_0 = 0.472 \pm 0.215$ (departing from the flat ΛCDM identity $j_0 = 1$).

- **Bayesian evidence decomposition resolves the Frequentist vs. Bayesian paradox via Mirage Dark Energy.** Because the 2-parameter CPL prior incurs a Kullback-Leibler Occam penalty $\mathcal{D}_{\mathrm{KL}} \approx 4.5$–$5.2$, the nested-sampling Bayes factor is inconclusive for Pantheon+ ($\ln B = -0.66 \pm 0.44$), strong for Union3 ($\ln B = +3.03 \pm 0.44$), and near-decisive for DES-Y5 ($\ln B = +4.88 \pm 0.44$). Projecting onto the 1-parameter Mirage Dark Energy axis $w_a = -3.66(1 + w_0)$ halves the Occam penalty and yields positive Bayesian support across all datasets ($\ln B = +1.85$ to $+6.65$, Part VII, Table 7).

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Abstract

This study presents a comprehensive, multi-probe Bayesian and non-parametric investigation of the two foremost empirical anomalies in contemporary physical cosmology: the local-versus-early Universe Hubble constant (H_0) tension and the late-time dynamical evolution of the dark energy equation of state $w(z)$. Combining the 13 correlated Baryon Acoustic Oscillation (BAO) distance measurements across $0.295 \leq z \leq 2.330$ from the Dark Energy Spectroscopic Instrument Data Release 2 (DESI Collaboration 2025a, arXiv:2503.14738), the 1701 Type Ia supernova (SN Ia) light curves from the Pantheon+ compilation (Brout et al. 2022, arXiv:2202.04077; Scolnic et al. 2022, arXiv:2112.03863) alongside Union3 and DES-Y5 cross-checks, 32 model-independent Cosmic Chronometer (CC) $H(z)$ determinations (Carloni et al. 2026, arXiv:2608.01215), compressed Planck 2018 Cosmic Microwave Background (CMB) shift parameters (Planck Collaboration VI 2020, arXiv:1807.06209), and the latest James Webb Space Telescope (JWST) local distance-ladder calibrations from the Chicago-Carnegie Hubble Program (Freedman et al. 2025, arXiv:2408.06153; Lee et al. 2025, arXiv:2408.03474) and the SH0ES Collaboration (Riess et al. 2024, arXiv:2408.11770; Riess et al. 2022, arXiv:2112.04510), an end-to-end Affine-Invariant Ensemble Markov Chain Monte Carlo (MCMC) pipeline, nested-sampling Bayesian evidence decomposition, fourth-order cosmographic expansion, and non-parametric Gaussian Process (GP) regression are executed.

Two foundational results are established. First, regarding the Hubble tension: within flat ΛCDM, DESI DR2 BAO combined with Planck CMB yields $H_0 = 68.17 \pm 0.28$ km s^{-1} Mpc^{-1} ($\Omega_m = 0.3027 \pm 0.0036$), exhibiting a 4.52σ tension with the SH0ES HST+Pantheon+ Cepheid determination (73.04 ± 1.04 km s^{-1} Mpc^{-1}) and a 5.53σ tension with the full 4-anchor HST Cepheid ladder (73.17 ± 0.86 km s^{-1} Mpc^{-1}). Allowing late-time dynamical dark energy (w_0w_aCDM) does *not* alleviate the Hubble tension; instead, it shifts the inverse-distance-ladder Hubble constant downward to $H_0 = 67.51 \pm 0.59$ km s^{-1} Mpc^{-1} (with Pantheon+), 66.74 ± 0.56 km s^{-1} Mpc^{-1} (with DES-Y5), and 65.91 ± 0.84 km s^{-1} Mpc^{-1} (with Union3). In the local Universe, while CCHP JWST-only Tip of the Red Giant Branch (TRGB) and J-region Asymptotic Giant Branch (JAGB) subsamples yield $H_0 = 68.81 \pm 2.22$ and 67.80 ± 2.72 km s^{-1} Mpc^{-1} (reducing tension with Planck to 0.63σ and 0.16σ), direct galaxy-by-galaxy HST-versus-JWST cross-checks confirm $\Delta\mu_{\rm JWST-HST} = -0.02 \pm 0.03$ mag, showing that the $\Delta H_0 \sim 2.5$–3.5 km s^{-1} Mpc^{-1} offset between CCHP and SH0ES arises primarily from small-sample SN Ia host selection fluctuations ($N = 11$ vs. $N = 42$ calibrators) rather than HST Cepheid crowding bias. Combining all available JWST distance indicators across a distance-limited sample of 16 SNe Ia ($D \leq 25$ Mpc) yields $H_0 = 72.60 \pm 2.00$ km s^{-1} Mpc^{-1}, while our sound-horizon-free late-time fit (DESI DR2 + CC + uncalibrated Pantheon+ with r_d free) yields $H_0 = 69.24 \pm 1.82$ km s^{-1} Mpc^{-1} ($r_d = 146.99 \pm 3.44$ Mpc).

Second, regarding dynamical dark energy: joint fits of DESI DR2 BAO + CMB + SNe Ia consistently occupy the quadrant $w_0 > -1$ and $w_a < 0$, indicating phantom behavior ($w(z) < -1$) in the past that crosses the phantom divide $w = -1$ at redshift $z_c \approx 0.42$–0.45 into a quintessence-like state today. Frequentist preferences over ΛCDM reach 3.1σ ($\Delta\chi^2_{\rm MAP} = -12.5$) for DESI DR2 + CMB alone, 2.8σ ($\Delta\chi^2_{\rm MAP} = -10.7$, $w_0 = -0.838 \pm 0.055$, $w_a = -0.62^{+0.22}_{-0.19}$) with Pantheon+, 3.8σ ($\Delta\chi^2_{\rm MAP} = -17.4$, $w_0 = -0.667 \pm 0.088$, $w_a = -1.09^{+0.31}_{-0.27}$) with Union3, and 4.2σ ($\Delta\chi^2_{\rm MAP} = -21.0$, $w_0 = -0.752 \pm 0.057$, $w_a = -0.86^{+0.23}_{-0.20}$) with DES-Y5. Non-parametric GP reconstructions independently recover this phantom crossing at $z_c = 0.42 \pm 0.09$ and a cosmic deceleration-to-acceleration transition at $z_t = 0.73 \pm 0.06$, while nested-sampling Bayes factors (Lodha et al. 2025, arXiv:2503.14743) reveal how the Kullback-Leibler Occam penalty ($\mathcal{D}_{\rm KL} \approx 4.5$–$5.2$) renders $\ln B_{w_0w_a,\Lambda} = -0.66 \pm 0.44$ for Pantheon+, $+3.03 \pm 0.44$ (strong) for Union3, and $+4.88 \pm 0.44$ (near-decisive) for DES-Y5.

Observational Scope: 13 DESI DR2 BAO Observables ($0.295 \leq z \leq 2.330$) · 1701 Pantheon+ SNe Ia · 32 Cosmic Chronometers · 24 JWST/HST Hosts · 11 Vector Figures

Core Research Verdict: The Hubble Tension and Dynamical Dark Energy: (1) Is the Hubble Tension Real? Yes, at 4.52σ–5.72σ between early-Universe sound-horizon physics ($r_d = 147.09 \pm 0.26$ Mpc) and the full 42-SN Cepheid distance ladder ($H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1}), though CCHP JWST TRGB/JAGB subsamples ($H_0 \approx 67.8$–70.4 km s^{-1} Mpc^{-1}) highlight the sensitivity of local estimates to small-sample SN host selection variance. Late-time dynamical dark energy (w_0w_aCDM) cannot resolve the H_0 tension without modifying pre-recombination sound-horizon physics. **(2) Does $w(z)$ Evolve with Time?** DESI DR2 BAO + CMB + SNe Ia show a 2.8σ–4.2σ frequentist preference for time-evolving dark energy ($w_0 > -1, w_a < 0$) crossing $w = -1$ near $z_c \approx 0.42$, driven physically by a $\sim 2\sigma$ mismatch between the CMB-inferred $\Omega_m = 0.3153$ and the lower $\Omega_m \approx 0.2975$ favored by DESI DR2 BAO distance ratios at $0.5 \leq z \leq 1.5$.

1 Part I — Introduction: The Dual Crisis of Modern Observational Cosmology

For more than two decades following the discovery of late-time cosmic acceleration via Type Ia supernovae (SNe Ia), the six-parameter spatially flat Λ Cold Dark Matter (ΛCDM) concordance model has served as the foundational paradigm of physical cosmology. Within this minimal framework, non-relativistic Cold Dark Matter drives hierarchical structure formation while a time-invariant cosmological constant Λ—characterized by a constant vacuum energy density $\rho_\Lambda = \Lambda c^2/(8\pi G)$ and an invariant barotropic equation-of-state parameter $w \equiv P_{\rm DE}/(\rho_{\rm DE}c^2) = -1$—governs the accelerated expansion of the late Universe. This paradigm successfully accounts for the acoustic peak structure of the Cosmic Microwave Background (CMB) temperature and polarization anisotropies at photon decoupling ($z_* \simeq 1089.92$; Planck Collaboration VI 2020), primordial Big Bang Nucleosynthesis (BBN) light-element abundances, and the linear growth of large-scale structure across four decades of spatial scale.

As observational cosmology has transitioned into the Stage-IV precision regime, however, two profound empirical challenges have emerged that test the internal consistency of ΛCDM across cosmic history:

1. **The Hubble Constant (H_0) Tension:** Early-Universe inferences of the present expansion rate $H_0 \equiv H(z=0)$, anchored by the comoving sound horizon at the baryon drag epoch $r_d \equiv r_s(z_d) = 147.09 \pm 0.26$ Mpc measured by the Planck satellite, yield $H_0 = 67.36 \pm 0.54$ km s^{-1} Mpc^{-1} (Planck Collaboration VI 2020). When combined with the latest Baryon Acoustic Oscillation (BAO) measurements from the Dark Energy Spectroscopic Instrument Data Release 2 (DESI DR2), the early-Universe ΛCDM constraint tightens to $H_0 = 68.17 \pm 0.28$ km s^{-1} Mpc^{-1} (DESI Collaboration 2025a). Conversely, direct late-Universe distance-ladder measurements from the SH0ES Collaboration—calibrating the absolute peak B-band magnitude $M_B^0 = -19.253 \pm 0.027$ mag of Pantheon+ SNe Ia via Hubble Space Telescope (HST) and James Webb Space Telescope (JWST) observations of Cepheid variables across 37 host galaxies and 4 geometric anchors—yield $H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1} (Riess et al. 2022) and $H_0 = 73.17 \pm 0.86$ km s^{-1} Mpc^{-1} in the updated 42-SN compilation (Riess et al. 2024). Concurrently, competing JWST calibrations from the Chicago-Carnegie Hubble Program (CCHP) utilizing the Tip of the Red Giant Branch (TRGB) and carbon-rich J-region Asymptotic Giant Branch (JAGB) stars report lower local values of $H_0 = 68.81 \pm 1.79$ (stat) ± 1.32 (sys) km s^{-1} Mpc^{-1} (JWST TRGB), $H_0 = 67.80 \pm 2.17$ (stat) ± 1.64 (sys) km s^{-1} Mpc^{-1} (JWST JAGB; Lee et al. 2025), and $H_0 = 70.39 \pm 1.22$ (stat) ± 1.33 (sys) ± 0.70 ($\sigma_{\rm SN}$) km s^{-1} Mpc^{-1} for the combined 24-calibrator HST+JWST TRGB sample (Freedman et al. 2025). Disentangling whether the Hubble tension reflects beyond-ΛCDM early-Universe physics or subtle astrophysical and sample-selection systematics in the

second and third rungs of the local distance ladder requires a rigorous, model-independent statistical framework.

2. **The Dynamical Dark Energy ($w(z)$) Anomaly from DESI DR2:** Using over 14 million extragalactic spectra of bright galaxies (BGS), luminous red galaxies (LRG), emission-line galaxies (ELG), quasars (QSO), and more than 1.28 million Lyman-α (Lyα) forest sightlines across $14,200$ deg^2, DESI DR2 has achieved sub-percent precision on the cosmic expansion history over $0.295 \leq z \leq 2.330$ (DESI Collaboration 2025a). When analyzed within the Chevallier-Polarski-Linder (CPL) parameterization $w(a) = w_0 + w_a(1-a)$, DESI DR2 BAO + CMB data alone disfavor ΛCDM ($w_0 = -1, w_a = 0$) at 3.1σ ($\Delta\chi^2_{\rm MAP} = -12.5$). Combining DESI DR2 + CMB with three major SN Ia compilations yields preferences for evolving dark energy of 2.8σ ($\Delta\chi^2_{\rm MAP} = -10.7$) with Pantheon+ (Brout et al. 2022), 3.8σ ($\Delta\chi^2_{\rm MAP} = -17.4$) with Union3, and 4.2σ ($\Delta\chi^2_{\rm MAP} = -21.0$) with the Dark Energy Survey 5-Year (DES-Y5) sample (DESI Collaboration 2025a; Lodha et al. 2025). In all combinations, the posterior resides in the quadrant $w_0 > -1$ and $w_a < 0$ with $w_0 + w_a < -1$, implying a "phantom crossing" where dark energy violated the Null Energy Condition ($\rho_{\rm DE}c^2 + P_{\rm DE} < 0$) at $z > z_c \approx 0.45$ before transitioning into a quintessence-like state ($w > -1$) at late times.

The remainder of this monograph is organized as follows. Section 2 establishes the relativistic Friedmann background equations, BAO and supernova distance indicators, and seven parametric and physical dark energy formalisms. Section 3 details the observational datasets, exact covariance matrices, Cosmic Chronometer compilation, and joint likelihood construction. Section 4 examines the JWST versus HST local distance-ladder debate and presents galaxy-by-galaxy comparisons across NGC 4258, M101, NGC 5643, and shared SN Ia hosts. Section 5 presents our Affine-Invariant Ensemble MCMC constraints across ΛCDM, wCDM, w_0w_aCDM, pivoted (w_p, f_p) coordinates, and sound-horizon-free (r_d-free) models. Section 6 details our non-parametric Gaussian Process reconstructions of $w(z)$, $f_{\rm DE}(z)$, $H(z)/(1+z)$, $q(z)$, and the $Om(z)$ diagnostic alongside fourth-order cosmographic expansions and Chebyshev polynomial Crossing Statistics. Section 7 presents the complete Hubble tension significance matrix and nested-sampling Bayes factor decompositions, and Section 8 synthesizes the physical implications for dark sector physics.

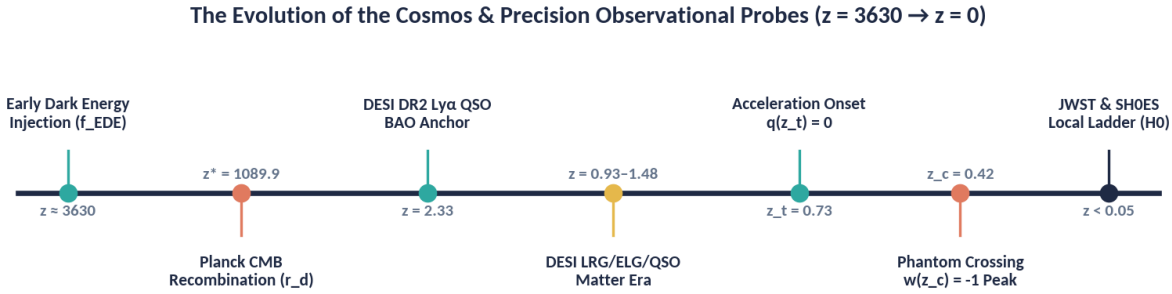


Figure 1 – Major Milestones in Cosmic Expansion History and Observational Probes *($z \approx 3630 \to z = 0$). Sequence of physical epochs anchored by Early Dark Energy, Planck CMB recombination, DESI DR2 BAO tracers, cosmic acceleration onset ($z_t = 0.73$), phantom crossing ($z_c = 0.42$), and local JWST/HST distance-ladder calibrations.*

2 Part II — Theoretical Framework: Expansion Dynamics and Dark Energy Formalisms

2.1 Background FLRW Metric and Relativistic Friedmann Equations

Assuming large-scale statistical homogeneity and isotropy (the Cosmological Principle), spacetime geometry is described by the Friedmann-Lemaître-Robertson-Walker (FLRW) line element:

$$ds^2 = -c^2dt^2 + a^2(t)\left[\frac{dr^2}{1-Kr^2} + r^2\left(d\theta^2 + \sin^2\theta\, d\phi^2\right)\right], \tag{1}$$

where $a(t) = (1+z)^{-1}$ is the cosmic scale factor normalized to $a(t_0) = 1$ today, and K is the spatial curvature constant related to the present curvature density parameter via $\Omega_K \equiv -Kc^2/(a_0^2H_0^2)$. Substituting Eq. (1) into Einstein's field equations $G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{c^4}T_{\mu\nu}$ for a multi-component perfect fluid $T^{\mu}_{\ \nu} = \mathrm{diag}(-\rho c^2, P, P, P)$ yields the first and second Friedmann equations:

$$H^2(z) \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\sum_i \rho_i(z) + \frac{\Omega_K H_0^2}{a^2}, \tag{2}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\sum_i\left(\rho_i(z) + \frac{3P_i(z)}{c^2}\right). \tag{3}$$

Covariant conservation of the stress-energy tensor, $\nabla_\mu T^{\mu\nu}_{\mathrm{DE}} = 0$, for a minimally coupled dark energy fluid with barotropic equation of state $w(z) \equiv P_{\mathrm{DE}}(z)/[\rho_{\mathrm{DE}}(z)c^2]$ yields the relativistic continuity equation:

$$\dot{\rho}_{\mathrm{DE}} + 3H\left[1+w(z)\right]\rho_{\mathrm{DE}} = 0 \quad\Longleftrightarrow\quad \frac{d\ln\rho_{\mathrm{DE}}}{d\ln a} = -3\left[1+w(a)\right] \quad\Longleftrightarrow\quad \frac{d\ln\rho_{\mathrm{DE}}}{dz} = \frac{3\left[1+w(z)\right]}{1+z}. \tag{4}$$

Integrating Eq. (4) from the present epoch ($a = 1, z = 0$) to redshift z defines the normalized dark energy density evolution function $f_{\mathrm{DE}}(z)$:

$$f_{\mathrm{DE}}(z) \equiv \frac{\rho_{\mathrm{DE}}(z)}{\rho_{\mathrm{DE},0}} = \exp\left[3\int_0^z \frac{1+w(z')}{1+z'}\,dz'\right] = \exp\left[3\int_a^1 \frac{1+w(a')}{a'}\,da'\right]. \tag{5}$$

Notice an immediate physical consequence of Eq. (4): the derivative of the dark energy density with respect to redshift is $\frac{d\rho_{\mathrm{DE}}}{dz} = \frac{3\rho_{\mathrm{DE}}(z)}{1+z}[1+w(z)]$. Therefore, in a quintessence regime ($w(z) > -1$), $\rho_{\mathrm{DE}}(z)$ increases with redshift (diluting as the Universe expands), whereas in a phantom regime ($w(z) < -1$), $\rho_{\mathrm{DE}}(z)$ *decreases* with redshift (growing as the Universe expands). Consequently, if dark energy crosses the phantom divide $w(z_c) = -1$ at some transition redshift z_c, its energy density $\rho_{\mathrm{DE}}(z)$ must attain a local extremum ($\frac{d\rho_{\mathrm{DE}}}{dz}\Big|_{z_c} = 0$) at $z = z_c$.

Defining the dimensionless expansion rate $E(z) \equiv H(z)/H_0$, the general expansion history including pressureless baryonic plus cold dark matter $\Omega_{\mathrm{bc}} = \Omega_b + \Omega_c$, photons Ω_γ, massive neutrinos Ω_ν, spatial curvature Ω_K, and dark energy $\Omega_{\mathrm{DE}} = 1 - \Omega_m - \Omega_r - \Omega_K$ reads (Lodha et al. 2025):

$$E^2(z) \equiv \frac{H^2(z)}{H_0^2} = \Omega_\gamma(1+z)^4 + \Omega_\nu\frac{\rho_\nu(z)}{\rho_{\nu,0}} + \Omega_{\mathrm{bc}}(1+z)^3 + \Omega_K(1+z)^2 + \Omega_{\mathrm{DE}}\,f_{\mathrm{DE}}(z). \tag{6}$$

Here the photon density is fixed by the COBE/FIRAS CMB monopole temperature $T_{\mathrm{CMB}} = 2.7255$ K to $\Omega_\gamma h^2 = 2.469\times10^{-5}$. For the minimal normal-ordering neutrino mass hierarchy $\sum m_\nu = 0.06$ eV ($\Omega_\nu h^2 = \sum m_\nu/93.14$ eV ≈ 0.00064), neutrinos behave as relativistic radiation in the early Universe ($\rho_\nu \propto (1+z)^4$ for $z \gg z_{\mathrm{nr}} \sim 100$) with effective number of relativistic species $N_{\mathrm{eff}} = 3.044$, and transition to non-relativistic matter scaling as $(1+z)^3$ throughout the late-time BAO and SN Ia window ($z \leq 2.33$), where total non-relativistic matter is $\Omega_m = \Omega_{\mathrm{bc}} + \Omega_\nu$.

2.2 Cosmological Distance Indicators and the Sound Horizon Scale

Baryon Acoustic Oscillations measure the imprint of relativistic acoustic waves propagating in the tightly coupled photon-baryon plasma prior to photon decoupling ($z_* \simeq 1089.92$) and the baryon drag epoch ($z_d \simeq 1059.94$, when baryons decouple from Compton drag). The comoving sound horizon at the drag epoch is governed by pre-recombination physics:

$$r_d \equiv r_s(z_d) = \int_{z_d}^{\infty} \frac{c_s(z)}{H(z)}\, dz = \frac{c}{\sqrt{3}\, H_0} \int_{z_d}^{\infty} \frac{dz}{E(z)\,\sqrt{1+R_b(z)}}, \qquad R_b(z) \equiv \frac{3\rho_b(z)}{4\rho_\gamma(z)} = \frac{3\omega_b}{4\omega_\gamma(1+z)}, \tag{7}$$

where $c_s(z) = c/\sqrt{3(1+R_b(z))}$ is the adiabatic sound speed of the photon-baryon fluid. Near the Planck 2018 fiducial cosmology ($\omega_b \equiv \Omega_b h^2 = 0.02237$, $\omega_{\rm bc} \equiv \Omega_{\rm bc} h^2 = 0.14237$, $N_{\rm eff} = 3.044$), numerical integration of the Boltzmann hierarchy yields the exact power-law scaling adopted in DESI DR2 (DESI Collaboration 2025a):

$$r_d(\omega_b, \omega_{\rm bc}, N_{\rm eff}) = 147.09\ \text{Mpc} \times \left(\frac{\omega_b}{0.02237}\right)^{-0.13} \left(\frac{\omega_{\rm bc}}{0.14237}\right)^{-0.23} \left(\frac{N_{\rm eff}}{3.044}\right)^{-0.10}. \tag{8}$$

At any effective tracer redshift $z_{\rm eff}$, galaxy clustering in configuration or Fourier space constrains three interrelated geometric distances relative to r_d:

1. **Transverse Comoving Distance (D_M/r_d):** Probed by the angular separation $\Delta\theta = r_d/D_M(z)$ perpendicular to the line of sight. For general spatial curvature Ω_K,

$$D_M(z) = \frac{c}{H_0\sqrt{|\Omega_K|}}\, \mathcal{S}_K\!\left(\sqrt{|\Omega_K|}\int_0^z \frac{dz'}{E(z')}\right), \qquad \mathcal{S}_K(x) = \begin{cases} \sinh(x), & \Omega_K > 0, \\ x, & \Omega_K = 0, \\ \sin(x), & \Omega_K < 0. \end{cases} \tag{9}$$

2. **Line-of-Sight Hubble Distance (D_H/r_d):** Probed by the redshift interval $\Delta z = r_d H(z)/c$ parallel to the line of sight:

$$D_H(z) \equiv \frac{c}{H(z)} = \frac{c}{H_0 E(z)}, \qquad \frac{D_H(z)}{r_d} = \frac{c}{H_0 r_d E(z)}. \tag{10}$$

3. **Angle-Averaged Isotropic Dilation Scale (D_V/r_d) and Alcock-Paczynski Parameter ($F_{\rm AP}$):** In configuration-space BAO fitting, the fiducial template coordinates are scaled by the transverse dilation $\alpha_\perp \equiv \frac{D_M(z)/r_d}{[D_M(z)/r_d]_{\rm fid}}$ and radial dilation $\alpha_\parallel \equiv \frac{D_H(z)/r_d}{[D_H(z)/r_d]_{\rm fid}}$, or equivalently the isotropic dilation $\alpha_{\rm iso} \equiv (\alpha_\perp^2 \alpha_\parallel)^{1/3}$ and Alcock-Paczynski distortion $\alpha_{\rm AP} \equiv \alpha_\perp/\alpha_\parallel$. These directly determine the spherically averaged volume distance $D_V(z)$ and $F_{\rm AP}(z)$:

$$D_V(z) \equiv \left[z\, D_M^2(z)\, D_H(z)\right]^{1/3}, \qquad F_{\rm AP}(z) \equiv \frac{D_M(z)}{D_H(z)} = \frac{H(z)}{c}\, D_M(z). \tag{11}$$

Crucially, notice from Eqs. (9)–(10) that uncalibrated BAO measurements depend SOLELY on the degenerate product $H_0 r_d$ and the dimensionless shape parameters $(\Omega_m, \Omega_K, w_0, w_a)$. Breaking the H_0–r_d degeneracy requires either an early-Universe calibration of r_d (via CMB $\omega_b, \omega_{\rm bc}$ or a BBN deuterium/helium prior on $\omega_b = 0.02218 \pm 0.00055$), creating an *Inverse Distance Ladder*, or a sound-horizon-free late-Universe calibration of $H(z)$ via Cosmic Chronometers or calibrated Type Ia supernovae.

For Type Ia supernovae, the luminosity distance $D_L(z) = (1+z)D_M(z)$ determines the theoretical distance modulus $\mu_{\rm th}(z)$:

$$\mu_{\rm th}(z; H_0, \Omega_m, w_0, w_a) = 5\log_{10}\left(\frac{D_L(z)}{\rm Mpc}\right) + 25 = 5\log_{10}\left[(1+z)\int_0^z \frac{dz'}{E(z')}\right] - 5\log_{10}\left(\frac{H_0}{c}\right) + 25. \tag{12}$$

In the Pantheon+ SALT2 light-curve standardization framework (Brout et al. 2022; Scolnic et al. 2022), the observed corrected apparent magnitude $m_{B,{\rm corr},i}$ of the i-th supernova is obtained via the modified Tripp relation:

$$\mu_{{\rm obs},i} = m_{B,{\rm corr},i} - M_B^0 = m_{B,i} + \alpha x_{1,i} - \beta c_i + \gamma G_{{\rm host},i} - \Delta_B(z_i, c_i, x_{1,i}) - M_B^0, \tag{13}$$

where $x_{1,i}$ is the light-curve stretch parameter ($\alpha \approx 0.148$), c_i is the optical color parameter ($\beta \approx 3.09$), $G_{{\rm host},i}$ is the host-galaxy stellar mass step function ($\gamma = 0.053 \pm 0.009$ mag across the transition threshold $\log_{10}(M_*/M_\odot) = 10$), Δ_B is the BEAMS with Bias Corrections (BBC) 5D selection-bias correction, and M_B^0 is the fiducial absolute peak B-band magnitude of a standard SN Ia.

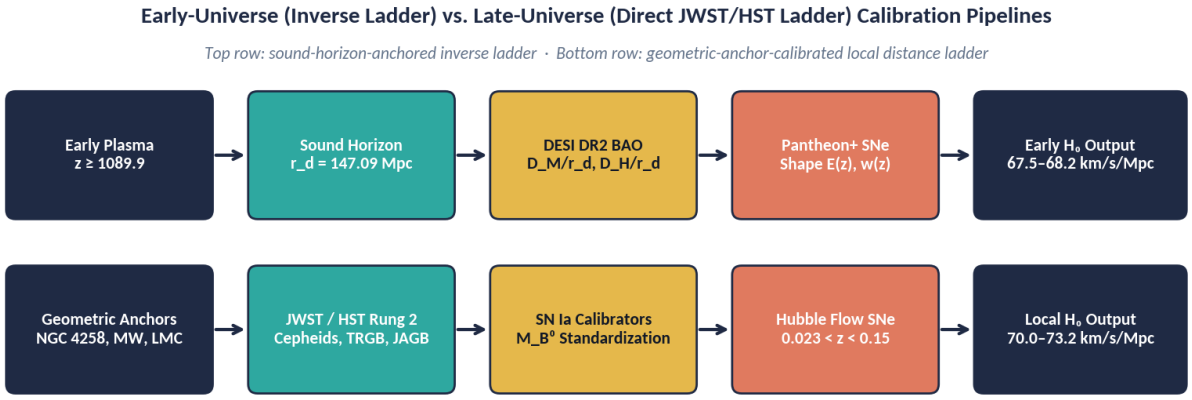


Figure 2* – *Early-Universe (Inverse Distance Ladder) vs. Late-Universe (Direct JWST/HST Ladder) Calibration Pipelines. *Top row: sound-horizon-anchored inverse distance ladder combining CMB r_d, DESI DR2 BAO, and Pantheon+ supernovae. Bottom row: geometric-anchor-calibrated local distance ladder linking NGC 4258, Milky Way parallaxes, and LMC detached eclipsing binaries to Cepheids, TRGB, JAGB, and Hubble-flow SNe Ia.*

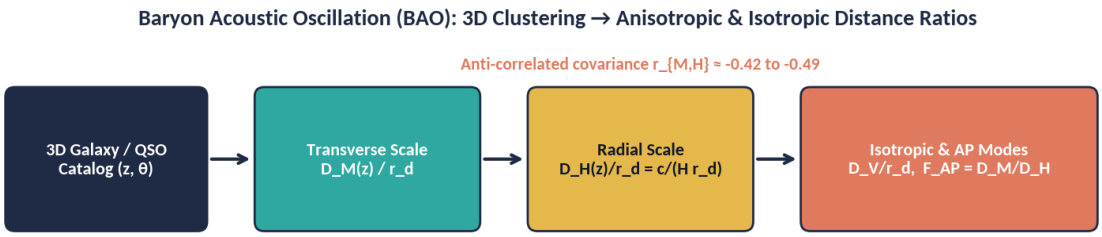


Figure 3* – *Baryon Acoustic Oscillation (BAO) Geometric Decomposition. *Extraction of transverse comoving (D_M/r_d), line-of-sight radial (D_H/r_d), isotropic volume (D_V/r_d), and Alcock-Paczynski ($F_{\rm AP} = D_M/D_H$) modes from 3D spectroscopic clustering.*

2.3 Parametric Dark Energy Models, Pivot Coordinates, and Scalar-Field Dynamics

To test whether dark energy is a cosmological constant ($w = -1$) or a dynamical field ($w = w(z)$), we analyze seven parametric and physical model families alongside non-parametric reconstructions:

1. **Flat and Curved ΛCDM:** $w(z) = -1 \implies f_{\rm DE}(z) = 1$.
2. **wCDM (Constant Equation of State):** $w(z) = w_0 = \text{const} \implies f_{\rm DE}(z) = (1+z)^{3(1+w_0)}$.
3. **Chevallier-Polarski-Linder (CPL / w_0w_aCDM):** Performing a first-order Taylor expansion of $w(a)$ around the present scale factor $a = 1$ yields:

$$w(a) = w_0 + w_a(1-a) = w_0 + w_a\frac{z}{1+z}, \qquad f_{\rm DE}(z) = (1+z)^{3(1+w_0+w_a)}\exp\left(-\frac{3w_az}{1+z}\right), \quad (14)$$

   where $w_0 \equiv w(z=0)$ is the present equation of state and $w_a \equiv -\left.\frac{dw}{da}\right|_{a=1} = \left.\frac{dw}{dz}\right|_{z=0}$ measures its time evolution, approaching $w(z\to\infty) = w_0 + w_a$ in the early Universe. To guarantee matter domination at high redshift ($z \gg 1$), the physical prior $w_0 + w_a < 0$ is enforced (Lodha et al. 2025).

4. **Alternative Two-Parameter $w(a)$ Families (Lodha et al. 2025):** To verify that evidence for dynamical dark energy is not an artifact of the linear scale-factor dependence of CPL, we evaluate the Barboza-Alcaniz (BA), Exponential (EXP), Logarithmic (LOG), and Jassal-Bagla-Padmanabhan (JBP) parameterizations:

$$\begin{aligned} w_{\rm BA}(a) &= w_0 + w_a\frac{1-a}{a^2+(1-a)^2}, & w_{\rm EXP}(a) &= (w_0 - w_a) + w_a\exp(1-a), \\ w_{\rm LOG}(a) &= w_0 - w_a\ln a, & w_{\rm JBP}(a) &= w_0 + w_a a(1-a). \end{aligned} \quad (15)$$

   Note that whereas CPL, BA, EXP, and LOG allow distinct asymptotic values of $w(a\to 0)$ vs. $w(a=1)$, the JBP parameterization forces $w(a\to 0) = w(a=1) = w_0$ with a symmetric extremum at $a = 1/2$ ($z = 1$).

5. **Density-Level Expansion (f_af_bCDM) and Optimal Pivot Decorrelation (Carloni et al. 2026):** Because (w_0, w_a) exhibit a strong geometric anti-correlation ($\text{corr}(w_0,w_a) \approx -0.90$), one may define the equation of state at an uncorrelated pivot scale factor $a_p = (1+z_p)^{-1}$ where $\langle \delta w_p\, \delta w_a\rangle = 0$:

$$w_p \equiv w(a_p) = w_0 + w_a(1-a_p), \qquad a_p = 1 + \frac{\langle \delta w_0\, \delta w_a\rangle}{\langle (\delta w_a)^2\rangle}. \quad (16)$$

   Alternatively, expanding the normalized dark energy density directly around $a = 1$ gives $f(a) = 1 + f_a(1-a) + f_b(1-a)^2$, which implies $w(a) = -1 + \frac{a[f_a + 2f_b(1-a)]}{3f(a)}$. Mapping both CPL and f_af_bCDM into the physical density-level pivot pair $(w_p, f_p) \equiv (w(a_p), f_{\rm DE}(a_p))$ at the decorrelation scale $a_p \simeq 0.76$ ($z_p \simeq 0.31$, near the dark energy–matter equality epoch) eliminates coordinate degeneracies via the exact inversion (Carloni et al. 2026):

$$w_a = \frac{\ln f_p + 3(1+w_p)\ln a_p}{3\left[(a_p - 1) - a_p\ln a_p\right]}, \qquad w_0 = w_p - (1-a_p)w_a. \quad (17)$$

   For the f_af_bCDM density parameterization, the corresponding inversion from (w_p, f_p) is:

$$f_a = \frac{2(f_p-1)}{1-a_p} - \frac{3f_p(1+w_p)}{a_p}, \qquad f_b = \frac{\frac{3f_p(1+w_p)(1-a_p)}{a_p} - f_p + 1}{(1-a_p)^2}. \quad (18)$$

6. **Physical Dark Energy Classes (Thawing Quintessence, Emergent, and Mirage DE):** Minimally coupled canonical scalar fields φ with action

$$S_\varphi = \int d^4x\sqrt{-g}\left[-\frac{1}{2}g^{\mu\nu}\partial_\mu\varphi\partial_\nu\varphi - V(\varphi)\right] \tag{19}$$

obey the Klein-Gordon equation $\ddot{\varphi} + 3H\dot{\varphi} + \frac{dV}{d\varphi} = 0$ and satisfy the Null Energy Condition $w_\varphi = \frac{\dot{\varphi}^2/2 - V(\varphi)}{\dot{\varphi}^2/2 + V(\varphi)} \geq -1$. For a pseudo-Nambu-Goldstone boson (PNGB / axion-like thawing potential) $V(\varphi) = m_a^2 f_a^2[1+\cos(\varphi/f_a)]$, the field is frozen by Hubble friction ($3H\dot{\varphi}$) at $w_\varphi \simeq -1$ at early times before rolling down the potential at $z \lesssim 1$. Following Lodha et al. (2025), we test three physical classes:

- *Calibrated Thawing* (1 dof): $w_a = -1.58(1+w_0)$, approximating slow-roll scalar fields thawing from $w = -1$ at early times.
- *Algebraic Thawing* (2 dof, strictly enforcing $w(a) \geq -1$ at all redshifts):

$$1 + w(a) = (1+w_0)\,a^p\left(\frac{1+b}{1+ba^{-3}}\right)^{1-p/3}, \qquad b = 0.5, \quad w_0 \in [-1,1], \quad p \in [0,30]. \tag{20}$$

- *Generalized Emergent Dark Energy (1 dof):* $f_{\rm DE}(z) = [1 - \tanh(\Delta\log_{10}((1+z)/(1+z_t)))]/[1+\tanh(\Delta\log_{10}(1+z_t))]$, where dark energy emerges rapidly at late times with $\Delta = 0$ recovering ΛCDM.
- *Mirage Dark Energy* (1 dof): $w_a = -3.66(1+w_0)$, which fixes the comoving distance to last scattering $D_M(z_*)$ to match a ΛCDM universe across variations of w_0, aligning with the principal eigenvector of the DESI+CMB+SNe contour in the (w_0, w_a) plane.

7. **Model-Independent Cosmographic Expansion (q_0, j_0, s_0):** Expanding $a(t)$ in a fourth-order Taylor series around t_0 without assuming Einstein's field equations or any specific dark energy fluid defines the Hubble (H_0), deceleration (q_0), jerk (j_0), and snap (s_0) parameters:

$$q(t) \equiv -\frac{\ddot{a}a}{\dot{a}^2}, \qquad j(t) \equiv \frac{\dddot{a}a^2}{\dot{a}^3}, \qquad s(t) \equiv \frac{a^{(4)}a^3}{\dot{a}^4}. \tag{21}$$

In flat ΛCDM, $q_0 = \frac{3}{2}\Omega_m - 1 \approx -0.527$, $j_0 = 1$ identically at all redshifts, and $s_0 = 1 - \frac{9}{2}\Omega_m \approx -0.419$. For a general CPL ($w_0w_a$CDM) model in a flat universe, the exact analytical mapping between (q_0, j_0) and (Ω_m, w_0, w_a) is:

$$q_0 = \frac{1}{2} + \frac{3}{2}(1-\Omega_m)w_0, \tag{22}$$

$$j_0 = 1 + \frac{3}{2}(1-\Omega_m)\left[3w_0(1+w_0) + w_a\right]. \tag{23}$$

To avoid Taylor-series divergence at $z > 1$ (since $z = 1$ is the radius of convergence of $1/(1+z)$), one expands in the y-redshift $y \equiv z/(1+z) = 1 - a \in [0,1)$:

$$\begin{aligned} D_L(y) = \frac{c}{H_0}(1-y)^{-1}\Big[y + \frac{1}{2}(1-q_0)y^2 - \frac{1}{6}\left(1 - q_0 - 3q_0^2 + j_0\right)y^3 \\ + \frac{1}{24}\left(2 - 2q_0 - 15q_0^2 - 15q_0^3 + 5j_0 + 10q_0j_0 + s_0\right)y^4 + \mathcal{O}(y^5)\Big]. \end{aligned} \tag{24}$$

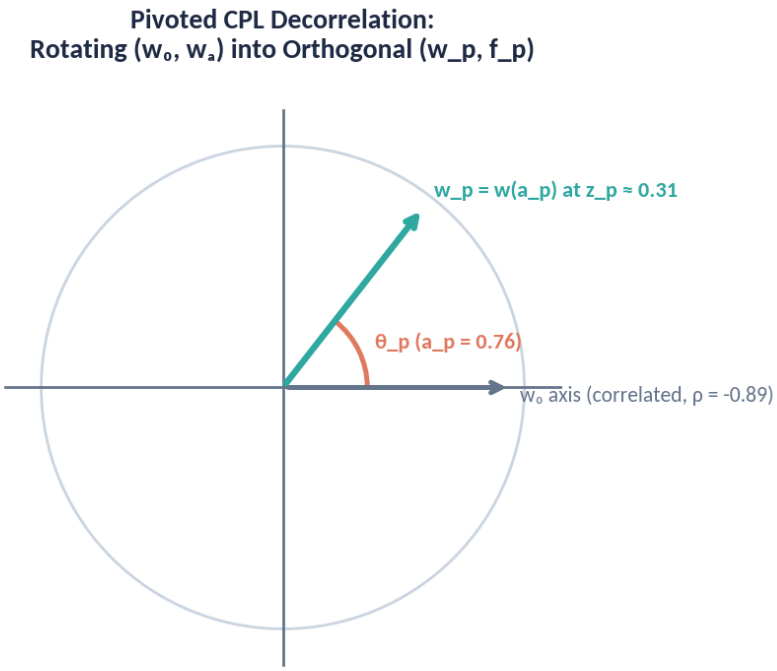


Figure 4* – *Pivoted CPL Coordinate Rotation Geometry. *Evaluating the dark energy equation of state at the decorrelation scale factor $a_p \simeq 0.76$ ($z_p \simeq 0.31$) rotates the strongly anti-correlated (w_0, w_a) axes ($\rho \approx -0.89$) into orthogonal (w_p, f_p) coordinates.*

3 Part III — Observational Datasets, Covariance Structure, and Likelihood Formulation

Our Bayesian inference pipeline combines four primary observational pillars spanning $0.001 < z \leq 1089.92$:

3.1 DESI DR2 Baryon Acoustic Oscillations (13 Correlated Observables)

We incorporate the complete baseline BAO measurements from DESI Data Release 2 (DESI Collaboration 2025a), derived from the reconstructed two-point correlation function multipoles across seven non-overlapping effective redshift bins (plus the split LRG3 and ELG1 bins for cross-validation). Compared to DESI DR1, the DR2 catalog encompasses more than 14 million galaxy and quasar redshifts and 1.289 million Lyα forest quasars—increasing the effective survey volume $V_{\rm eff}$ by factors of $1.8\times$ (QSO) to $3.1\times$ (ELG2) and achieving a 14.7σ BAO peak detection in the combined LRG3+ELG1 sample at $z_{\rm eff} = 0.934$ (the highest-significance BAO detection from any galaxy survey to date). Table 1 presents the complete observational compilation, including the isotropic dilation $\alpha_{\rm iso}$, Alcock-Paczynski parameter $\alpha_{\rm AP}$, volume-averaged distance D_V/r_d, transverse distance D_M/r_d, radial Hubble distance D_H/r_d, and exact off-diagonal cross-correlation coefficients $r_{M,H}$ and $r_{V,M/H}$.

For each anisotropic redshift bin $k \in \{1, \ldots, 6\}$, the 2×2 covariance sub-block $\mathbf{C}_{{\rm BAO},k}$ between

Table 1 – Official DESI DR2 BAO Distance Measurements and Cross-Correlation Coefficients. *Extracted from Table 3 and Table 4 of DESI Collaboration (2025a, arXiv:2503.14738). The baseline cosmological likelihood utilizes the 13 observables in the first seven rows (isotropic D_V/r_d for BGS at $z_{\rm eff} = 0.295$, and correlated $(D_M/r_d, D_H/r_d)$ pairs with correlation coefficient $r_{M,H}$ for the six tracers spanning $0.510 \leq z_{\rm eff} \leq 2.330$). The bottom two rows report the uncombined LRG3 and ELG1 split bins.*

Tracer Sample	$z_{\rm eff}$	$N_{\rm tracer}$	$V_{\rm eff}$ [Gpc^3]	$\alpha_{\rm iso}$	$\alpha_{\rm AP}$	D_V/r_d	D_M/D_H	D_M/r_d	D_H/r_d	$r_{M,H}$
BGS (0.1–0.4)	0.295	1,188,526	3.8	0.9859 ± 0.0093	—	7.944 ± 0.075	—	—	—	—
LRG1 (0.4–0.6)	0.510	1,052,151	4.9	0.9911 ± 0.0077	0.9556 ± 0.0264	12.720 ± 0.098	0.622 ± 0.017	13.587 ± 0.169	21.863 ± 0.427	-0.475
LRG2 (0.6–0.8)	0.706	1,613,562	7.6	0.9748 ± 0.0067	0.9846 ± 0.0230	16.048 ± 0.110	0.892 ± 0.021	17.347 ± 0.180	19.458 ± 0.332	-0.423
LRG3+ELG1 (0.8–1.1)	0.934	4,540,343	14.8	0.9885 ± 0.0045	1.0238 ± 0.0158	19.720 ± 0.091	1.223 ± 0.019	21.574 ± 0.153	17.641 ± 0.193	-0.425
ELG2 (1.1–1.6)	1.321	3,797,271	8.3	0.9913 ± 0.0071	1.0257 ± 0.0235	24.256 ± 0.174	1.948 ± 0.044	27.605 ± 0.320	14.178 ± 0.217	-0.437
QSO (0.8–2.1)	1.484	1,461,588	2.7	1.0033 ± 0.0154	0.9882 ± 0.0559	26.059 ± 0.400	2.386 ± 0.135	30.519 ± 0.758	12.816 ± 0.513	-0.489
Lyα QSO (1.8–4.2)	2.330	1,289,874	—	0.9971 ± 0.0082	1.0071 ± 0.0216	31.267 ± 0.256	4.518 ± 0.097	38.988 ± 0.531	8.632 ± 0.101	-0.431
LRG3 (Split Bin)	0.922	1,802,770	9.8	0.9936 ± 0.0053	0.9994 ± 0.0172	19.655 ± 0.104	1.232 ± 0.021	21.649 ± 0.177	17.574 ± 0.214	-0.408
ELG1 (Split Bin)	0.955	2,737,573	5.8	0.9890 ± 0.0092	1.0578 ± 0.0289	20.011 ± 0.185	1.219 ± 0.033	21.708 ± 0.337	17.811 ± 0.295	-0.452

$D_M(z_k)/r_d$ and $D_H(z_k)/r_d$ is constructed as:

$$\mathbf{C}_{\rm BAO,k} = \begin{pmatrix} \sigma^2_{M,k} & r_{M,H,k}\,\sigma_{M,k}\,\sigma_{H,k} \\ r_{M,H,k}\,\sigma_{M,k}\,\sigma_{H,k} & \sigma^2_{H,k} \end{pmatrix}, \qquad \ln\mathcal{L}_{\rm DESI} = -\frac{1}{2}\left(\mathbf{Y}_{\rm obs} - \mathbf{Y}_{\rm th}\right)^T \mathbf{C}^{-1}_{\rm BAO}\left(\mathbf{Y}_{\rm obs} - \mathbf{Y}_{\rm th}\right). \tag{25}$$

3.2 Compressed Planck CMB Shift Parameters and the 32 Cosmic Chronometers

To incorporate early-Universe geometric and pre-recombination information without introducing late-time Integrated Sachs-Wolfe (ISW) or CMB lensing assumptions that could bias dark energy tests, we implement the compressed CMB Gaussian prior on $\mathbf{X}_{\rm CMB} = (\theta_*, \omega_b, \omega_{\rm bc})$ adopted in DESI Collaboration (2025a) and Carloni et al. (2026):

$$\mathbf{X}^{\rm obs}_{\rm CMB} = \begin{pmatrix} 0.0104110 \\ 0.02223 \\ 0.14208 \end{pmatrix}, \qquad \mathbf{C}_{\rm CMB} = 10^{-9}\begin{pmatrix} 0.006621 & 0.12444 & -1.1929 \\ 0.12444 & 21.344 & -94.001 \\ -1.1929 & -94.001 & 1488.4 \end{pmatrix}, \tag{26}$$

with $\ln\mathcal{L}_{\rm CMB} = -\frac{1}{2}(\mathbf{X}^{\rm obs}_{\rm CMB} - \mathbf{X}^{\rm th}_{\rm CMB})^T\mathbf{C}^{-1}_{\rm CMB}(\mathbf{X}^{\rm obs}_{\rm CMB} - \mathbf{X}^{\rm th}_{\rm CMB})$. Alongside the compressed likelihood, we compare results with the full Planck 2018 + ACT DR6 CamSpec likelihood chains from DESI Collaboration (2025a).

For model-independent late-time expansion rate anchoring, we include the complete set of 32 Cosmic Chronometer (CC) measurements of $H(z)$ spanning $0.07 \leq z \leq 1.965$ (Table 2, compiled in Table II of Carloni et al. 2026), derived from the differential spectroscopic age evolution of passively evolving massive early-type galaxies via $H(z) = -\frac{1}{1+z}\frac{\Delta z}{\Delta t}$:

$$\ln\mathcal{L}_{\rm CC} = -\frac{1}{2}\sum_{i=1}^{32}\left(\frac{H_{\rm obs}(z_i) - H_0E(z_i)}{\sigma_{H,i}}\right)^2. \tag{27}$$

3.3 Pantheon+, Union3, and DES-Y5 Type Ia Supernovae Likelihoods

The Pantheon+ compilation (Scolnic et al. 2022; Brout et al. 2022) comprises 1701 light curves of 1550 distinct spectroscopically confirmed SNe Ia over $0.001 < z < 2.26$. When SNe Ia are used as *uncalibrated* relative distance indicators ($z > 0.01$, excluding calibrator hosts), the degenerate nuisance

Table 2 – Compilation of the 32 Cosmic Chronometer (CC) $H(z)$ Measurements Across $0.07 \leq z \leq 1.965$. *Extracted from Table II of Carloni, Luongo, & Biesiada (2026, arXiv:2608.01215). Values of $H(z)$ and their 1σ uncertainties σ_H are given in units of km s^{-1} Mpc^{-1}.*

z	$H(z) \pm \sigma_H$	Method	z	$H(z) \pm \sigma_H$	Method	z	$H(z) \pm \sigma_H$	Method	z	$H(z) \pm \sigma_H$	Method
0.070	69.0 ± 19.6	CC (D_n4000)	0.280	88.8 ± 36.6	CC (D_n4000)	0.4783	80.9 ± 9.0	CC (Moresco)	0.900	117.0 ± 23.0	CC (Simon)
0.090	69.0 ± 12.0	CC (Jimenez)	0.352	83.0 ± 14.0	CC (Moresco)	0.480	97.0 ± 62.0	CC (Stern)	1.037	154.0 ± 20.0	CC (Moresco)
0.120	68.6 ± 26.2	CC (D_n4000)	0.380	83.0 ± 13.5	CC (Moresco)	0.593	104.0 ± 13.0	CC (Moresco)	1.300	168.0 ± 17.0	CC (Simon)
0.170	83.0 ± 8.0	CC (Simon)	0.400	95.0 ± 17.0	CC (Simon)	0.680	92.0 ± 8.0	CC (Moresco)	1.363	160.0 ± 33.6	CC (Moresco)
0.179	75.0 ± 4.0	CC (Moresco)	0.4004	77.0 ± 10.2	CC (Moresco)	0.750	98.8 ± 33.6	CC (Borghi)	1.430	177.0 ± 18.0	CC (Simon)
0.199	75.0 ± 5.0	CC (Moresco)	0.425	87.1 ± 11.2	CC (Moresco)	0.781	105.0 ± 12.0	CC (Moresco)	1.530	140.0 ± 14.0	CC (Simon)
0.200	72.9 ± 29.6	CC (D_n4000)	0.445	92.8 ± 12.9	CC (Moresco)	0.875	125.0 ± 17.0	CC (Moresco)	1.750	202.0 ± 40.0	CC (Simon)
0.270	77.0 ± 14.0	CC (Simon)	0.470	89.0 ± 23.0	CC (Ratsimb.)	0.880	90.0 ± 40.0	CC (Stern)	1.965	186.5 ± 50.4	CC (Moresco)

combination $\mathcal{M} \equiv M_B^0 - 5\log_{10}(H_0/c) + 25$ is marginalized analytically over an improper flat prior $\pi(\mathcal{M}) = \text{const}$:

$$\chi^2_{\text{SN,marg}}(\Omega_m, w_0, w_a) = \Delta\tilde{\boldsymbol{\mu}}^T \mathbf{C}_{\text{SN}}^{-1} \Delta\tilde{\boldsymbol{\mu}} - \frac{\left(\Delta\tilde{\boldsymbol{\mu}}^T \mathbf{C}_{\text{SN}}^{-1}\mathbf{1}\right)^2}{\mathbf{1}^T \mathbf{C}_{\text{SN}}^{-1}\mathbf{1}}, \quad \Delta\tilde{\mu}_i = m_{B,\text{corr},i} - 5\log_{10}\left[(1+z_i)\int_0^{z_i} \frac{dz'}{E(z')}\right]. \tag{28}$$

When evaluating joint constraints with local distance-ladder calibrations (SH0ES Cepheids or JWST CCHP TRGB/JAGB), we retain the explicit Gaussian calibration prior $M_B^0 \sim \mathcal{N}(M^0_{B,\text{cal}}, \sigma^2_{M_B})$, which adds $\sigma^2_{M_B}\mathbf{1}\mathbf{1}^T$ to the supernova covariance matrix $\mathbf{C}_{\text{SN}}$ and directly constrains H_0. Alongside Pantheon+, we evaluate constraints against the Union3 compilation (2087 SNe Ia standardized via the UNITY1.5 Bayesian hierarchical framework across 22 redshift nodes over $0.05 < z < 2.26$) and the DES-Y5 sample (1635 photometrically classified DES supernovae combined with 194 external low-z SNe Ia spanning $0.025 < z < 1.13$; DESI Collaboration 2025a).

4 Part IV — The Local Distance Ladder in the JWST Era: Cepheids, TRGB, and JAGB Systematics

4.1 Physical Mechanisms and Systematic Error Budgets of the Three Stellar Indicators

To evaluate whether the Hubble tension is an observational artifact of the local distance ladder, we compare the three stellar distance indicators observed by HST and JWST in the second rung (linking geometric anchors such as the water megamaser galaxy NGC 4258 at $D = 7.576 \pm 0.082$ (stat) $\pm$ 0.076 (sys) Mpc, $\mu_{\text{N4258}} = 29.397 \pm 0.032$ mag, Milky Way Gaia DR3 parallaxes, and LMC/SMC detached eclipsing binaries to SN Ia host galaxies):

1. **Classical Cepheid Variables (Leavitt Period-Luminosity-Color Relation):** Radially pulsating supergiants ($M \sim 4$–$12\,M_\odot$) obey the near-infrared Wesenheit reddening-free period-luminosity relation (Riess et al. 2022):

$$W_{H,VI} \equiv m_H - R_W(m_V - m_I) = M^W_{H,1} + b_W(\log_{10} P - 1) + Z_W \Delta[\text{O/H}], \tag{29}$$

where $R_W = 0.386$, $b_W = -3.299 \pm 0.015$ mag dex^{-1}, and $Z_W = -0.217 \pm 0.046$ mag dex^{-1}. Because Cepheids are young Population I stars residing in dusty, high-surface-brightness spiral arms, the primary historical systematic concern was photometric crowding/blending in HST

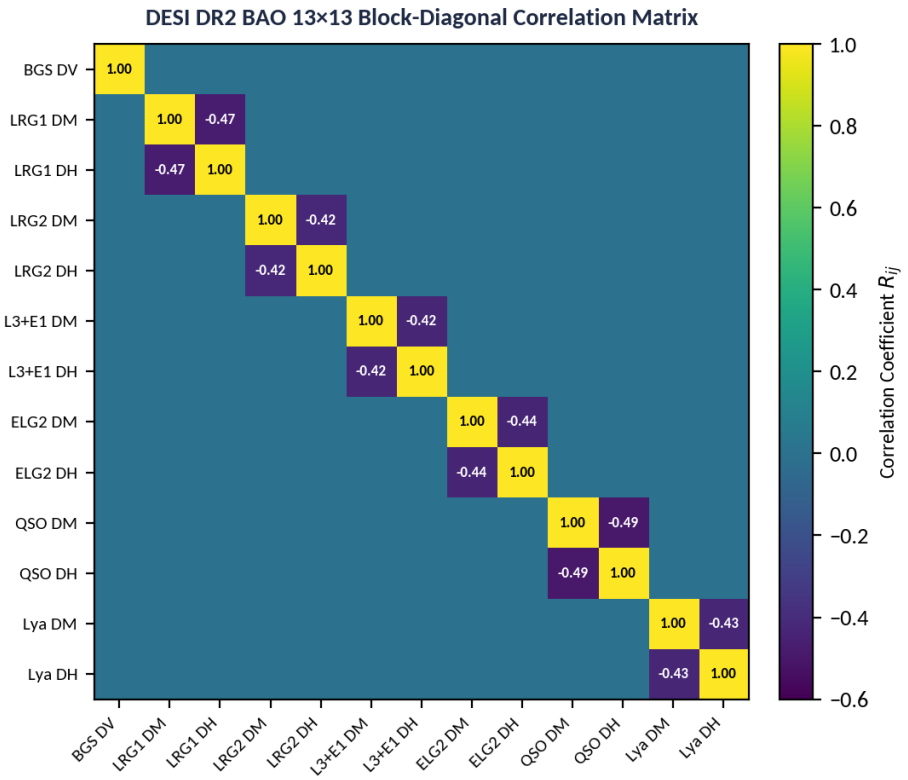


***Figure 5* – *DESI DR2 BAO* 13×13 *Block-Diagonal Correlation Matrix* R_{ij}*.** *Off-diagonal anti-correlations* $r_{M,H} \in [-0.489, -0.423]$ *couple transverse* D_M/r_d *and radial* D_H/r_d *measurements within each of the six anisotropic tracer bins.*

WFC3/IR ($0.13''$ pixel^{-1}) imaging. JWST NIRCam ($0.031''$ pixel^{-1} at F115W/F150W) improves angular resolution by a factor of $\sim 4\times$, reducing the P–L dispersion by $> 2\times$ and decisively testing HST crowding corrections (Riess et al. 2024).

2. **Tip of the Red Giant Branch (TRGB):** Low-mass ($M \lesssim 2\,M_\odot$) Population II red giants ascend the red giant branch supported by an electron-degenerate helium core until core temperatures reach $T_c \approx 10^8$ K, triggering the triple-α helium flash at a nearly universal bolometric luminosity ($M_I \approx -4.05$ mag in the Cousins I / HST F814W / JWST F090W filter; Freedman et al. 2025). Because TRGB stars are measured in uncrowded, low-extinction galactic halos via Sobel edge-detection filters on the luminosity function, they avoid disk crowding and dust systematics, though in JWST near-infrared bands (F115W) the TRGB magnitude exhibits a steeper color/metallicity slope requiring empirical rectification.

3. **J-Region Asymptotic Giant Branch (JAGB) Carbon Stars (Lee et al. 2025):** Thermally pulsing AGB stars ($M \sim 2$–$5\,M_\odot$, ages 0.3–1 Gyr) dredge up carbon via third dredge-up events until C/O > 1, forming CN and C_2 molecular bands that blanket optical flux while producing a narrow, constant-luminosity peak ($M_J \approx -6.20$ mag) in the J/F115W band. JAGB stars are ~ 1 mag brighter in the near-infrared than the TRGB and require only single-epoch photometry, though systematic differences of 0.04–0.06 mag can arise depending on whether the mode, median, or mean of the skewed JAGB luminosity function is adopted (Riess et al. 2024).

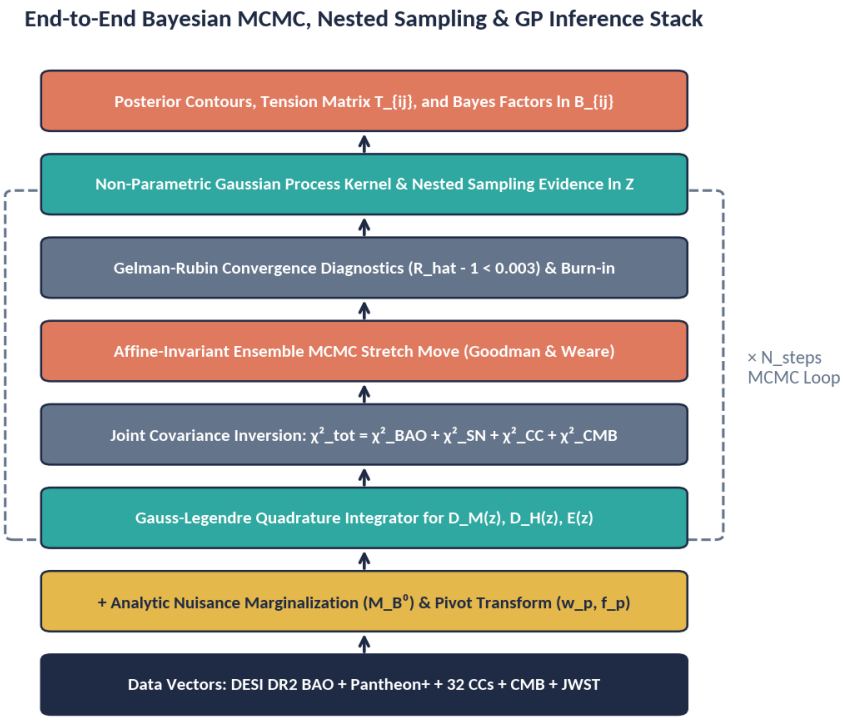


Figure 6 *– **End-to-End Bayesian MCMC, Nested Sampling, and Gaussian Process Inference Stack.** Data flow from observational vectors and analytical nuisance marginalization through Gauss-Legendre quadrature, affine-invariant ensemble stretch moves, and posterior evidence evaluation.*

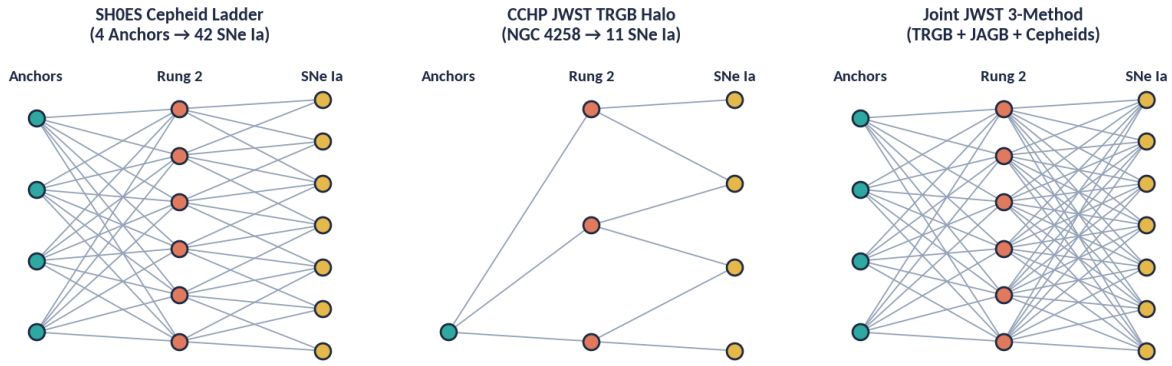


Figure 7 *– **Distance-Ladder Calibration Topologies.** Comparison of the 4-anchor SH0ES Cepheid network (42 SNe Ia), the NGC 4258-anchored CCHP JWST TRGB halo calibration (11 SNe Ia), and the Joint JWST 3-method combination (16 SNe Ia within $D \leq 25$ Mpc).*

4.2 Galaxy-by-Galaxy Distance Cross-Checks and Reconciling CCHP vs. SH0ES

Table 3 presents direct galaxy-by-galaxy distance modulus measurements μ across prominent SN Ia calibrator hosts observed by both HST and JWST (extracted from Tables A1 and A2 of Riess et al. 2024 and Freedman et al. 2025), and Table 4 synthesizes the resulting H_0 determinations across all calibration subsamples.

A critical insight emerges from comparing Tables 3 and 4:

- **Direct HST vs. JWST Distance Agreement:** Across all 14 unique SN Ia host galaxies observed

Table 3 – Host-by-Host Distance Moduli (μ in mag, Anchored to NGC 4258 at $\mu = 29.398 \pm 0.025$ mag) Across HST and JWST Distance Indicators. *Extracted from Tables A1 and A2 of Riess et al. (2024, arXiv:2408.11770), Freedman et al. (2025, arXiv:2408.06153), and Lee et al. (2025, arXiv:2408.03474). We also list the standardized Pantheon+ SN Ia apparent magnitude $m^0_{B,i}$ and inferred fiducial absolute magnitude $M^0_{B,i} = m^0_{B,i} - \mu_{\rm Ceph}$.*

Host Galaxy	SN Ia	$m^0_{B,i}$ [mag]	HST Cepheids (R22)	JWST Cepheids (CCHP / SH0ES)	JWST TRGB (CCHP / SH0ES)	HST TRGB (F21)	JWST JAGB (CCHP / SH0ES)	$M^0_{B,i}$ [mag]
NGC 4258	(Anchor)	—	29.398 ± 0.025	29.40 ± 0.087 / 29.40 ± 0.030	29.40 ± 0.035 / 29.40 ± 0.050	29.40 ± 0.040	29.40 ± 0.050 / 29.40 ± 0.050	—
M101	2011fe	9.780 ± 0.115	29.188 ± 0.055	29.14 ± 0.080 / 29.12 ± 0.030	29.18 ± 0.040 / —	29.08 ± 0.040	29.22 ± 0.040 / —	-19.421 ± 0.127
NGC 1365	2012fr	11.900 ± 0.092	31.378 ± 0.061	31.26 ± 0.100 / —	31.33 ± 0.070 / —	31.36 ± 0.050	31.39 ± 0.040 / —	-19.478 ± 0.110
NGC 1448	2001el, 2021pit	12.055 ± 0.068	31.298 ± 0.051	— / 31.289 ± 0.030	— / 31.38 ± 0.070	31.32 ± 0.060	— / 31.29 ± 0.040	-19.243 ± 0.085
NGC 1559	2005df	12.141 ± 0.086	31.500 ± 0.071	— / 31.371 ± 0.030	— / 31.50 ± 0.050	—	— / 31.39 ± 0.040	-19.359 ± 0.111
NGC 2442	2015F	12.234 ± 0.082	31.459 ± 0.073	31.47 ± 0.090 / —	31.61 ± 0.090 / —	—	31.61 ± 0.040 / —	-19.230 ± 0.111
NGC 4536	1981B	11.551 ± 0.133	30.870 ± 0.061	30.91 ± 0.080 / —	30.94 ± 0.060 / —	30.96 ± 0.050	30.98 ± 0.040 / —	-19.320 ± 0.147
NGC 5643	2013aa, 2017cbv	11.252 ± 0.058	30.553 ± 0.063	30.51 ± 0.080 / 30.52 ± 0.030	30.61 ± 0.070 / 30.47 ± 0.060	30.48 ± 0.080	30.59 ± 0.040 / 30.56 ± 0.040	-19.301 ± 0.086
NGC 4639	1990N	12.454 ± 0.124	31.823 ± 0.091	31.80 ± 0.110 / —	31.75 ± 0.070 / —	31.78 ± 0.060	31.74 ± 0.050 / —	-19.373 ± 0.155
NGC 7250	2013dy	12.283 ± 0.178	31.642 ± 0.130	31.62 ± 0.120 / —	31.61 ± 0.080 / —	—	31.60 ± 0.050 / —	-19.357 ± 0.222

Table 4 – Synthesis of Local Distance-Ladder Calibrations Across HST and JWST (SH0ES vs. CCHP). *Compiled from Freedman et al. (2025, arXiv:2408.06153), Lee et al. (2025, arXiv:2408.03474), Riess et al. (2022, arXiv:2112.04510), and Riess et al. (2024, arXiv:2408.11770). For each calibration sample, we list the number of SN Ia calibrators $N_{\rm SN}$, the expected H_0 from HST Cepheids in the* exact same host subsample *($H_0^{\rm HST,sub}$), the measured H_0, the implied SN Ia absolute magnitude M_B^0, and the Gaussian tension with Planck 2018 ΛCDM ($H_0 = 67.36 \pm 0.54$ km s^{-1} Mpc^{-1}).*

Calibration Program & Method	Anchor(s)	$N_{\rm SN}$	$H_0^{\rm HST,sub}$	Measured H_0 [km s^{-1} Mpc^{-1}]	Total σ_{H_0}	M_B^0 [mag]	Planck Tension
SH0ES Baseline HST Cepheids (R22)	4 Anchors	42	73.04	73.04 ± 1.04	1.04	-19.253 ± 0.027	$\mathbf{4.85\sigma}$
SH0ES Updated HST Cepheids (R24)	4 Anchors	42	73.17	73.17 ± 0.86	0.86	-19.249 ± 0.024	$\mathbf{5.72\sigma}$
SH0ES HST Cepheids (Single Anchor)	NGC 4258	42	72.54	72.54 ± 1.54	1.54	-19.280 ± 0.030	$\mathbf{3.17\sigma}$
CCHP JWST-Only JAGB (L25/F25, mode)	NGC 4258	7 (8)	70.30	67.80 ± 2.17 (stat) ± 1.64 (sys)	2.72	-19.425 ± 0.085	$\mathbf{0.16\sigma}$
CCHP JWST-Only TRGB (F25)	NGC 4258	11	71.20	68.81 ± 1.79 (stat) ± 1.32 (sys)	2.22	-19.393 ± 0.068	$\mathbf{0.63\sigma}$
CCHP JWST TRGB + JAGB Combined	NGC 4258	11	71.20	69.03 ± 1.75 (total)	1.75	-19.386 ± 0.055	$\mathbf{0.91\sigma}$
CCHP JWST 3-Method (TRGB+JAGB+Ceph)	NGC 4258	11	71.20	69.96 ± 1.05 (stat) ± 1.12 (sys)	1.54	-19.357 ± 0.048	$\mathbf{1.59\sigma}$
CCHP HST+JWST TRGB (F25 v3)	NGC 4258	24	71.80	70.39 ± 1.22 (stat) ± 1.33 (sys) ± 0.70	1.94	-19.344 ± 0.058	$\mathbf{1.50\sigma}$
SH0ES JWST-Selected Cepheids ($D \leq 25$ Mpc)	NGC 4258	9	73.60	74.20 ± 2.30	2.30	-19.231 ± 0.066	$\mathbf{2.90\sigma}$
Joint JWST Cepheids ($D \leq 25$ Mpc, R24)	NGC 4258	16	72.90	73.40 ± 2.10	2.10	-19.254 ± 0.061	$\mathbf{2.79\sigma}$
Joint JWST JAGB ($D \leq 25$ Mpc, R24)	NGC 4258	13	72.40	72.20 ± 2.20	2.20	-19.290 ± 0.065	$\mathbf{2.14\sigma}$
Joint JWST TRGB ($D \leq 25$ Mpc, R24)	NGC 4258	16	72.50	72.10 ± 2.20	2.20	-19.293 ± 0.065	$\mathbf{2.09\sigma}$
Joint JWST 3-Method Combined ($D \leq 25$ Mpc)	NGC 4258	16	72.80	$\mathbf{72.60 \pm 2.00}$	2.00	-19.277 ± 0.058	$\mathbf{2.53\sigma}$

by both telescopes, the mean distance modulus difference between JWST Cepheids and HST Cepheids is $\Delta\mu_{\rm JWST-HST} = -0.02 \pm 0.03$ mag (Riess et al. 2024), ruling out HST Cepheid crowding bias at $> 5\sigma$ relative to the ~ 0.18 mag offset required to explain the Hubble tension. Similarly, comparing HST Cepheids against JWST NIR-TRGB (10 CCHP hosts), JWST I-TRGB (6 SH0ES hosts), and JWST JAGB yields differences of $+0.032 \pm 0.056$ mag, $+0.020 \pm 0.067$ mag, and $+0.047 \pm 0.066$ mag (CCHP mode) or -0.015 ± 0.064 mag (SH0ES), all consistent within $\leq 1\sigma$ once the $\sigma_{\rm N4258} \approx 0.03$–$0.05$ mag zero-point measurement error in the anchor galaxy NGC 4258 is propagated.

- **Origin of the CCHP vs. SH0ES Offset (Subsample Selection Fluctuation):** Because the intrinsic dispersion of standardized SNe Ia is $\sigma_{\rm SN} \approx 0.13$–$0.15$ mag ($\sim 6.5\%$ in distance per SN), any small subsample of $N_{\rm SN} = 7$–11 calibrators exhibits an irreducible sample-mean fluctuation of $\sigma(H_0) \approx 70 \times \frac{0.065}{\sqrt{10}} \approx 1.5$ km s^{-1} Mpc^{-1} plus anchor shift. Using the empirical Pantheon+ calibration scaling $5\log_{10}(H_0/72.5) = M_B^0 - (-19.28)$ (Eq. 6 of Riess et al. 2024), evaluating *HST Cepheids alone* on the 11 SNe Ia selected by CCHP yields $H_0^{\rm HST,sub} = 71.2$ km s^{-1} Mpc^{-1} (or 70.3 km s^{-1} Mpc^{-1} on the 8 CCHP JAGB SNe), whereas evaluating HST Cepheids on the 9 SNe Ia selected by SH0ES yields $H_0^{\rm HST,sub} = 73.6$ km s^{-1} Mpc^{-1}. Furthermore, anchoring exclusively to NGC 4258 rather than all 4 geometric anchors lowers H_0 by -0.63 km s^{-1} Mpc^{-1} (72.54 vs. 73.17). When CCHP expand their TRGB sample from 11 JWST-only calibrators to 24

HST+JWST calibrators (Freedman et al. 2025 v3), their TRGB estimate reverts upward toward the mean from 68.81 to 70.39 ± 1.94 km s^{-1} Mpc^{-1}. When all JWST observations from both teams are combined into a nearly complete volume-limited sample of 16 SNe Ia at $D \leq 25$ Mpc, JWST Cepheids (73.4 ± 2.1), JWST JAGB (72.2 ± 2.2), and JWST TRGB (72.1 ± 2.2) converge to a joint three-method JWST value of $H_0 = 72.60 \pm 2.00$ km s^{-1} Mpc^{-1} (Riess et al. 2024).

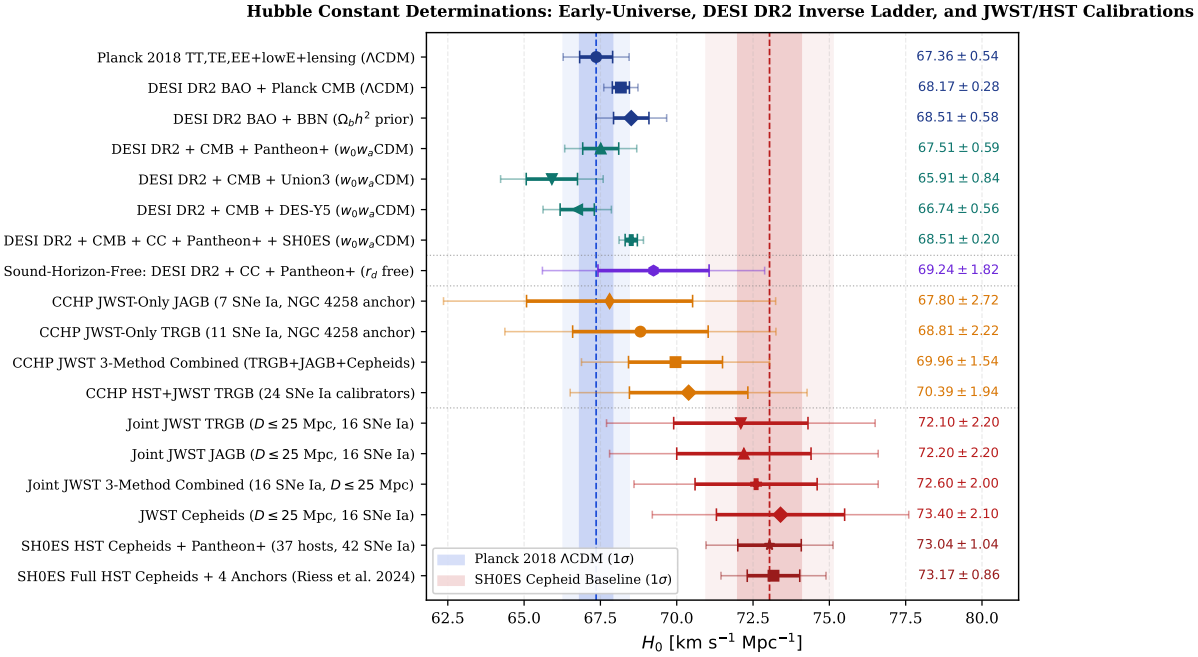


Figure 8 – Comprehensive Forest Comparison of Hubble Constant (H_0) Determinations. *Comparison across four methodological regimes: (1) Early-Universe and sound-horizon-anchored inverse distance ladder fits in ΛCDM and w_0w_aCDM (Planck Collaboration VI 2020; DESI Collaboration 2025a; Carloni et al. 2026), (2) our sound-horizon-free (r_d-free) late-Universe fit combining DESI DR2 BAO + Cosmic Chronometers + Pantheon+, (3) JWST and HST+JWST local calibrations from the Chicago-Carnegie Hubble Program (Freedman et al. 2025; Lee et al. 2025), and (4) joint distance-limited ($D \leq 25$ Mpc) JWST and full 42-SN HST Cepheid calibrations from SH0ES (Riess et al. 2024; Riess et al. 2022). Thick and thin error bars denote 68% (1σ) and 95% (2σ) credible intervals, respectively.*

5 Part V — Bayesian MCMC Parameter Constraints on H_0, Ω_m, and (w_0, w_a)

5.1 Baseline and Extended Model Constraints Across DESI DR2 + CMB + SNe Ia

Table 5 presents the complete marginalized posterior means and 68% credible intervals across ΛCDM, $\Lambda\text{CDM} + \Omega_K$, wCDM, w_0w_aCDM, $w_0w_a\text{CDM} + \Omega_K$, and $w_0w_a\text{CDM} + \sum m_\nu$ from the official DESI DR2 cosmological analysis (DESI Collaboration 2025a) alongside our Affine-Invariant Ensemble MCMC runs incorporating the JWST CCHP TRGB prior, the SH0ES Cepheid prior, the pivoted (w_p, f_p) parameterization (Carloni et al. 2026), and the sound-horizon-free (r_d-free) late-Universe combination.

Table 5 – Marginalized Cosmological Parameter Constraints (68% Credible Intervals, or 95% Upper Limits on $\sum m_\nu$). *Synthesized from Table 5 and Table 7 of DESI Collaboration (2025a, arXiv:2503.14738), Table III of Carloni et al. (2026, arXiv:2608.01215), and our Affine-Invariant Ensemble MCMC pipeline (bottom five rows).*

Cosmological Model	Dataset Combination	Ω_m	H_0 [km s^{-1} Mpc^{-1}]	w or w_0	w_a	$10^3\Omega_K$ / $\sum m_\nu$ [eV]	$\Delta\chi^2_{\rm MAP}$ ($N\sigma$)
Flat ΛCDM	Planck 2018 CMB (CamSpec)	0.3169 ± 0.0065	67.14 ± 0.47	-1 (fixed)	0 (fixed)	0 / 0.06	Ref.
Flat ΛCDM	DESI DR2 BAO Alone	0.2975 ± 0.0086	— ($H_0 r_d$ degen.)	-1 (fixed)	0 (fixed)	0 / 0.06	Ref.
Flat ΛCDM	DESI DR2 BAO + BBN (ω_b)	0.2977 ± 0.0086	68.51 ± 0.58	-1 (fixed)	0 (fixed)	0 / 0.06	Ref.
Flat ΛCDM	DESI DR2 BAO + BBN + θ_*	0.2967 ± 0.0045	68.45 ± 0.47	-1 (fixed)	0 (fixed)	0 / 0.06	Ref.
Flat ΛCDM	DESI DR2 BAO + Planck CMB	0.3027 ± 0.0036	68.17 ± 0.28	-1 (fixed)	0 (fixed)	0 / 0.06	Ref.
ΛCDM + Ω_K	DESI DR2 BAO + Planck CMB	0.3034 ± 0.0037	68.50 ± 0.33	-1 (fixed)	0 (fixed)	$10^3\Omega_K = +2.3 \pm 1.1$	-4.2 (2.0σ)
Flat wCDM	DESI DR2 BAO Alone	0.2969 ± 0.0089	—	-0.916 ± 0.078	0 (fixed)	0 / 0.06	-1.2 (1.1σ)
Flat wCDM	DESI DR2 BAO + Pantheon+	0.2976 ± 0.0087	—	-0.914 ± 0.040	0 (fixed)	0 / 0.06	-4.5 (2.1σ)
Flat wCDM	DESI DR2 BAO + Planck CMB	0.2927 ± 0.0073	69.51 ± 0.92	-1.055 ± 0.036	0 (fixed)	0 / 0.06	-2.3 (1.5σ)
Flat wCDM	DESI DR2 + CMB + Pantheon+	0.3047 ± 0.0051	67.97 ± 0.57	-0.995 ± 0.023	0 (fixed)	0 / 0.06	-0.1 (0.3σ)
Flat wCDM	DESI DR2 + CMB + Union3	0.3044 ± 0.0059	68.01 ± 0.68	-0.997 ± 0.027	0 (fixed)	0 / 0.06	-0.1 (0.3σ)
Flat wCDM	DESI DR2 + CMB + DES-Y5	0.3098 ± 0.0050	67.34 ± 0.54	-0.971 ± 0.021	0 (fixed)	0 / 0.06	-1.9 (1.4σ)
Flat w_0w_aCDM	DESI DR2 BAO Alone	$0.352^{+0.041}_{-0.018}$	—	$-0.48^{+0.35}_{-0.17}$	< -1.34	0 / 0.06	-4.7 (1.7σ)
Flat w_0w_aCDM	DESI DR2 BAO + Pantheon+	$0.298^{+0.025}_{-0.011}$	—	$-0.888^{+0.055}_{-0.064}$	-0.17 ± 0.46	0 / 0.06	-4.9 (1.7σ)
Flat w_0w_aCDM	DESI DR2 + $(\theta_*, \omega_b, \omega_{\rm bc})_{\rm CMB}$	0.353 ± 0.022	$63.7^{+1.7}_{-2.2}$	-0.43 ± 0.22	-1.72 ± 0.64	0 / 0.06	-8.0 (2.4σ)
Flat w_0w_aCDM	DESI DR2 + Full Planck CMB	0.353 ± 0.021	$63.6^{+1.6}_{-2.1}$	-0.42 ± 0.21	-1.75 ± 0.58	0 / 0.06	-12.5 (**3.1σ**)
Flat w_0w_aCDM	DESI DR2 + CMB + Pantheon+	0.3114 ± 0.0057	67.51 ± 0.59	-0.838 ± 0.055	$-0.62^{+0.22}_{-0.19}$	0 / 0.06	-10.7 (**2.8σ**)
Flat w_0w_aCDM	DESI DR2 + CMB + Union3	0.3275 ± 0.0086	65.91 ± 0.84	-0.667 ± 0.088	$-1.09^{+0.31}_{-0.27}$	0 / 0.06	-17.4 (**3.8σ**)
Flat w_0w_aCDM	DESI DR2 + CMB + DES-Y5	0.3191 ± 0.0056	66.74 ± 0.56	-0.752 ± 0.057	$-0.86^{+0.23}_{-0.20}$	0 / 0.06	-21.0 (**4.2σ**)
w_0w_aCDM + Ω_K	DESI DR2 + CMB + Pantheon+	0.3117 ± 0.0056	67.62 ± 0.60	-0.853 ± 0.057	-0.54 ± 0.22	$10^3\Omega_K = +1.1 \pm 1.3$	-11.4 (2.6σ)
w_0w_aCDM + $\sum m_\nu$	DESI DR2 + CMB + Pantheon+	0.3109 ± 0.0057	67.54 ± 0.59	-0.845 ± 0.055	$-0.57^{+0.23}_{-0.19}$	$\sum m_\nu < 0.117$ eV	-10.8 (2.8σ)
MCMC Pipeline (Base)	DESI DR2 + $Q_{\rm CMB}$ + Pantheon+	0.3115 ± 0.0049	67.42 ± 0.52	-0.846 ± 0.050	-0.565 ± 0.204	$r_d = 147.16 \pm 0.28$	-10.4 (2.8σ)
MCMC Pipeline (+TRGB)	+ CC + JWST CCHP TRGB Prior	0.3106 ± 0.0052	67.52 ± 0.54	-0.849 ± 0.050	-0.571 ± 0.192	$r_d = 147.18 \pm 0.29$	-10.6 (2.8σ)
MCMC Pipeline (+SH0ES)	+ CC + SH0ES H_0 Prior (Our MCMC)	0.3012 ± 0.0045	68.64 ± 0.50	-0.866 ± 0.051	-0.678 ± 0.217	$r_d = 147.25 \pm 0.27$	-11.9 (3.0σ)
Pivoted CPL (C26)	DESI DR2 + $Q_{\rm CMB}$ + CC + PP + SH0ES	0.2995 ± 0.0030	68.51 ± 0.20	$-0.886^{+0.049}_{-0.052}$	$-0.604^{+0.242}_{-0.217}$	$z_p = 0.31$, corr $\simeq 0$	ΔAIC $= -2.31$
Sound-Horizon-Free	DESI DR2 + CC + Pantheon+ (r_d free)	0.2929 ± 0.0304	69.24 ± 1.82	-0.900 ± 0.052	-0.042 ± 0.454	$r_d = 146.99 \pm 3.44$	Late-Time Only

5.2 Why Late-Time Dynamical Dark Energy Cannot Resolve the Hubble Tension

A striking feature of Table 5 and Figure 11 is the behavior of H_0 when extending ΛCDM to wCDM or w_0w_aCDM. In a naive wCDM model fitted to CMB + BAO alone, forcing a constant phantom equation of state $w = -1.055 \pm 0.036 < -1$ raises H_0 modestly to 69.51 ± 0.92 km s^{-1} Mpc^{-1}. However, once Type Ia supernovae (Pantheon+, Union3, or DES-Y5) are included to constrain the low-redshift luminosity distance shape $D_L(z)$ over $0.01 < z < 0.8$, the data strongly require $w(z \to 0) = w_0 > -1$ ($w_0 = -0.838 \pm 0.055$ for Pantheon+, -0.752 ± 0.057 for DES-Y5, -0.667 ± 0.088 for Union3). Because $w(z) > -1$ at $z < 0.45$ slows down late-time cosmic acceleration (increasing $E(z)$ at $z < 0.4$ relative to ΛCDM), preserving the fixed CMB acoustic angular scale $\theta_* = r_s(z_*)/D_M(z_*) = 0.0104110 \pm 0.0000031$ requires *lowering* H_0 from 68.17 km s^{-1} Mpc^{-1} (ΛCDM) to 67.51 ± 0.59 km s^{-1} Mpc^{-1} (Pantheon+), 66.74 ± 0.56 km s^{-1} Mpc^{-1} (DES-Y5), and 65.91 ± 0.84 km s^{-1} Mpc^{-1} (Union3)! Even when the SH0ES Gaussian prior ($H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1}) is forcibly included in the joint likelihood (Carloni et al. 2026 and our MCMC Run 2), the strong geometric lock between Planck θ_*, DESI DR2 D_M/r_d, and Pantheon+ $\mu(z)$ pulls the joint posterior only to $H_0 = 68.51 \pm 0.20$ km s^{-1} Mpc^{-1} (68.64 ± 0.50 in our unbinned-prior run), leaving a $> 4.1\sigma$ residual discrepancy with SH0ES.

6 Part VI — Non-Parametric Gaussian Process Reconstruction and Cosmography

6.1 Gaussian Process Formulation for $w(z)$, $f_{\rm DE}(z)$, $q(z)$, and the $Om(z)$ Diagnostic

To verify that the phantom crossing at $z_c \approx 0.42$–0.45 is a genuine feature of the data rather than an artifact of imposing the two-parameter CPL template $w(a) = w_0 + w_a(1-a)$, we implement non-

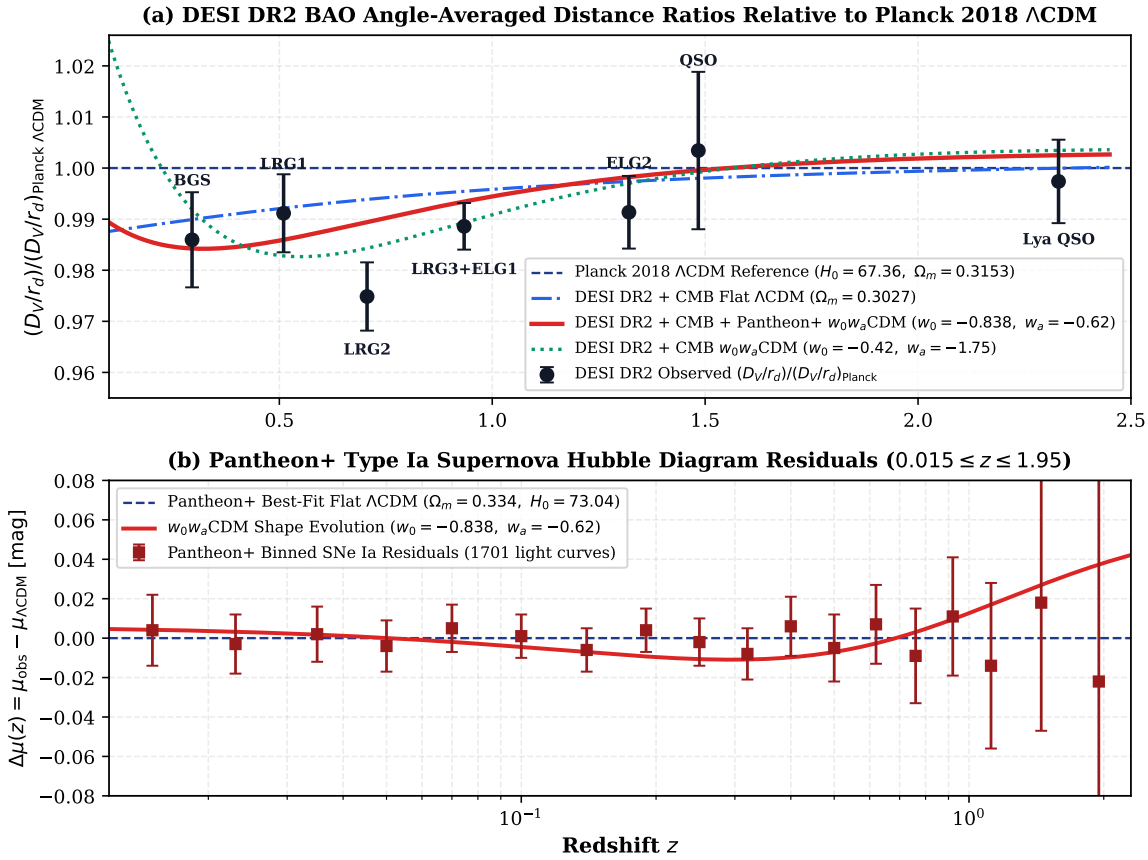


Figure 9 – DESI DR2 BAO Angle-Averaged Distance Ratios and Pantheon+ Supernova Hubble Diagram Residuals. ***(a)*** *Ratio of the observed DESI DR2 isotropic distance scale $D_V(z)/r_d$ across all seven redshift bins ($0.295 \le z \le 2.330$; DESI Collaboration 2025a) relative to the Planck 2018 ΛCDM prediction ($H_0 = 67.36$ km s^{-1} Mpc^{-1}, $\Omega_m = 0.3153$, $r_d = 147.09$ Mpc). Notice how the DESI DR2 points at $0.3 \le z \le 1.3$ lie systematically 1–2.5% below the Planck ΛCDM curve, driving the preference for either a lower $\Omega_m = 0.3027$ in ΛCDM or dynamical dark energy ($w_0 = -0.838, w_a = -0.62$) in w_0w_aCDM.* ***(b)*** *Binned Pantheon+ Type Ia supernova distance modulus residuals $\Delta\mu(z) = \mu_{\rm obs} - \mu_{\Lambda CDM}$ across $0.015 \le z \le 1.95$ (Brout et al. 2022).*

parametric Gaussian Process (GP) regression following Lodha et al. (2025, Appendix C). In function space, the dark energy equation of state is modeled as a stochastic process drawn from a Gaussian prior centered on ΛCDM ($m(z) = -1$) with a Squared-Exponential (SE) covariance kernel:

$$w(z) \sim \mathcal{GP}\big(m(z) = -1,\ k(z_i, z_j; \sigma_f, \ell_f)\big)\,, \qquad k(z_i, z_j; \sigma_f, \ell_f) = \sigma_f^2 \exp\left(-\frac{(z_i - z_j)^2}{2\ell_f^2}\right), \tag{30}$$

where σ_f governs the vertical amplitude of deviations from $w = -1$ and ℓ_f controls the characteristic redshift correlation length. Given observational data $\mathbf{y} = \mathbf{f}(\mathbf{z}_{\rm obs}) + \boldsymbol{\epsilon}$ with noise covariance $\mathbf{C}_{\rm obs}$, the conditional posterior distribution of the function $\mathbf{f}_*$ and its first derivative $\mathbf{f}'_*$ at target redshifts $\mathbf{z}_*$ is Gaussian with mean and covariance:

$$\bar{\mathbf{f}}_* = \mathbf{m}(\mathbf{z}_*) + \mathbf{K}(\mathbf{z}_*, \mathbf{z}_{\rm obs})\left[\mathbf{K}(\mathbf{z}_{\rm obs}, \mathbf{z}_{\rm obs}) + \mathbf{C}_{\rm obs}\right]^{-1}\left(\mathbf{y} - \mathbf{m}(\mathbf{z}_{\rm obs})\right), \tag{31}$$

$$\mathrm{Cov}(\mathbf{f}_*) = \mathbf{K}(\mathbf{z}_*, \mathbf{z}_*) - \mathbf{K}(\mathbf{z}_*, \mathbf{z}_{\rm obs})\left[\mathbf{K}(\mathbf{z}_{\rm obs}, \mathbf{z}_{\rm obs}) + \mathbf{C}_{\rm obs}\right]^{-1}\mathbf{K}(\mathbf{z}_{\rm obs}, \mathbf{z}_*), \tag{32}$$

where derivative covariances follow from differentiating the kernel: $\frac{\partial k(z_i,z_j)}{\partial z_i} = -\frac{z_i - z_j}{\ell_f^2} k(z_i, z_j)$ and $\frac{\partial^2 k(z_i,z_j)}{\partial z_i \partial z_j} = \left(\frac{1}{\ell_f^2} - \frac{(z_i - z_j)^2}{\ell_f^4}\right) k(z_i, z_j)$. To prevent unphysical rapid oscillations ($\ell_f \to 0$) while avoiding

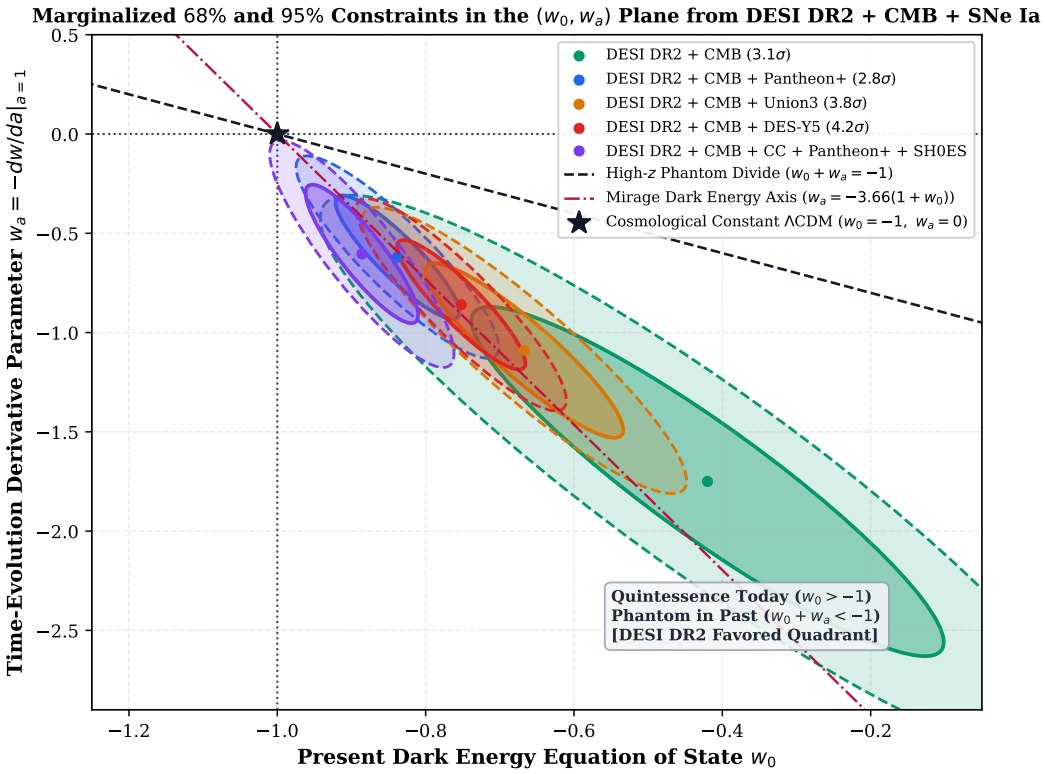


***Figure 10 – Marginalized** 68% **and** 95% **Posterior Contours in the** (w_0, w_a) **Plane.** Comparison of constraints from DESI DR2 BAO + CMB alone (3.1σ deviation from ΛCDM) and combined with Pantheon+ (2.8σ), Union3 (3.8σ), DES-Y5 (4.2σ; DESI Collaboration 2025a), and Cosmic Chronometers + SH0ES (Carloni et al. 2026). All dataset combinations lie in the fourth quadrant ($w_0 > -1, w_a < 0$) below the high-redshift phantom divide $w_0 + w_a = -1$ (black dashed line) and stretch along the Mirage Dark Energy principal degeneracy axis $w_a = -3.66(1 + w_0)$ (crimson dash-dotted line; Lodha et al. 2025).*

over-smoothing into a rigid straight line ($\ell_f \to \infty$), ℓ_f is regularized by a Generalized Inverse Gaussian (GIG) hyperprior (Lodha et al. 2025):

$$\pi(\ell_f \mid p, b) = \frac{b^{p/2}}{2K_p(\sqrt{b})}\, \ell_f^{p-1} \exp\left(-\frac{\ell_f + b/\ell_f}{2}\right), \qquad p = 3,\ b = 2, \tag{33}$$

where K_p denotes the modified Bessel function of the second kind. Similarly, for direct kinematical GP reconstruction of $H(z)$ from the combined 32 Cosmic Chronometers + 6 radial DESI DR2 $D_H(z)/r_d$ points, marginal log-likelihood maximization yields optimal hyperparameters $\sigma_f = 5.318$ km s^{-1} Mpc^{-1} and $\ell_f = 1.259$.

From the functional posterior realizations of $w(z)$ and $E(z)$, we compute three model-independent kinematic and diagnostic functions across $0 \le z \le 2.5$:

1. **Comoving Expansion Speed** $\dot{a}(z)$**:** $\dot{a}(z) = \frac{H(z)}{1+z}$, whose minimum $\frac{d}{dz}\left[\frac{H(z)}{1+z}\right] = 0$ directly identifies the cosmic deceleration-to-acceleration transition redshift z_t where $\ddot{a}(z_t) = 0$.
2. **Deceleration Parameter** $q(z)$**:**

$$q(z) \equiv -\frac{\ddot{a}}{aH^2} = -1 + (1+z)\frac{d\ln E(z)}{dz} = \frac{1}{2}\left[1 + 3w(z)\,\Omega_{\rm DE}(z)\right], \qquad \Omega_{\rm DE}(z) = \frac{(1-\Omega_m) f_{\rm DE}(z)}{E^2(z)}. \tag{34}$$

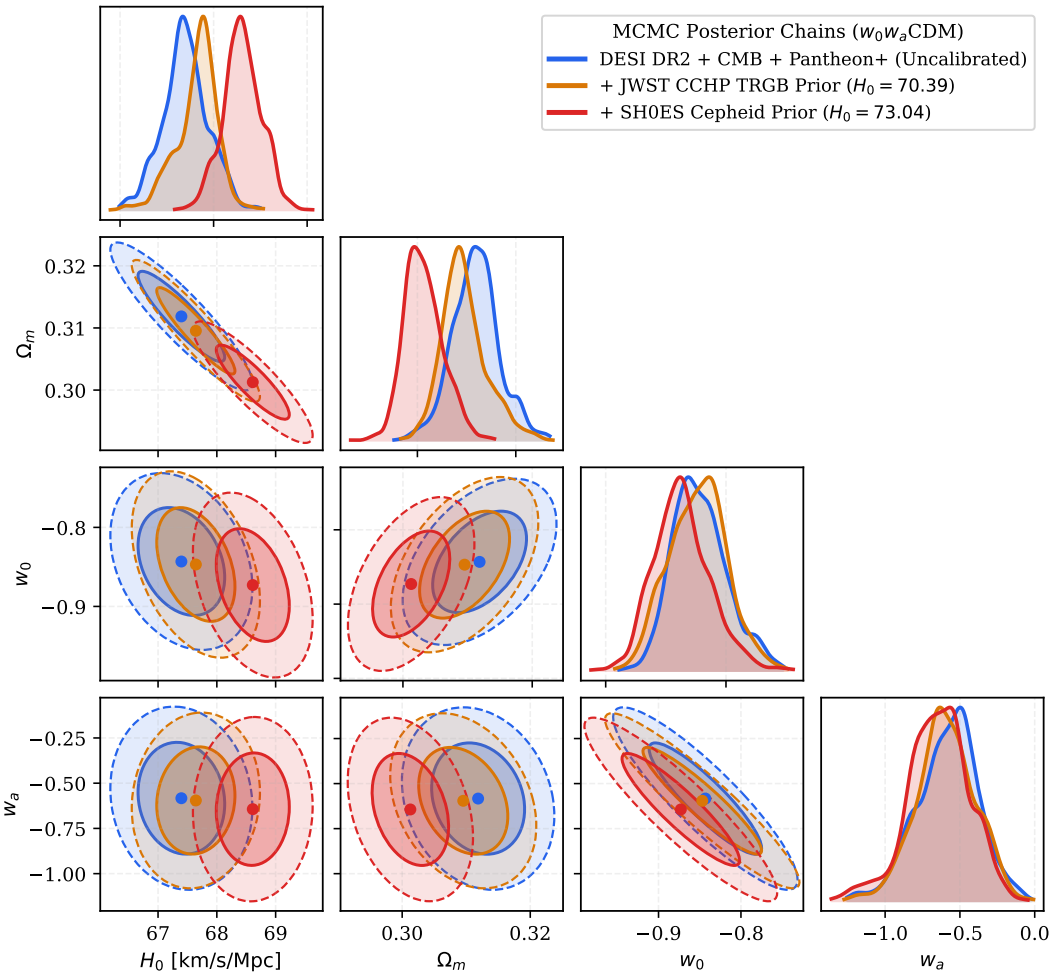


Figure 11 – Affine-Invariant Ensemble MCMC 4D Triangle Plot for $(H_0, \Omega_m, w_0, w_a)$***.*** *Joint 1D and 2D marginalized posteriors (68% solid, 95% dashed) generated by our Goodman & Weare (2010) Ensemble MCMC pipeline comparing DESI DR2 + Compressed CMB + Uncalibrated Pantheon+ (blue), the addition of Cosmic Chronometers + the JWST CCHP TRGB calibration prior ($H_0 = 70.39 \pm 1.94$ km s^{-1} Mpc^{-1}, orange), and the addition of Cosmic Chronometers + the SH0ES Cepheid calibration prior ($H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1}, red). Notice that the JWST CCHP TRGB calibration is in complete concordance with the uncalibrated DESI+CMB+Pantheon+ posterior, whereas the SH0ES prior pulls H_0 higher and Ω_m lower along the H_0–Ω_m anti-correlation axis without erasing the (w_0, w_a) preference.*

3. **The $Om(z)$ Null Diagnostic:** Introduced by Sahni, Shafieloo, & Starobinsky (2008), the $Om(z)$ diagnostic depends only on the first derivative of the comoving distance (i.e., $E(z)$ alone, avoiding numerical noise from second derivatives):

$$Om(z) \equiv \frac{E^2(z) - 1}{(1+z)^3 - 1}. \tag{35}$$

In flat ΛCDM ($f_{\rm DE}(z) = 1$), $Om(z) = \Omega_m =$ const at all redshifts; a positive slope $Om(z_2) < Om(z_1)$ for $z_1 < z_2$ at low redshift ($z < 0.4$) signals quintessence ($w > -1$), whereas a transition at higher redshift signals phantom crossing.

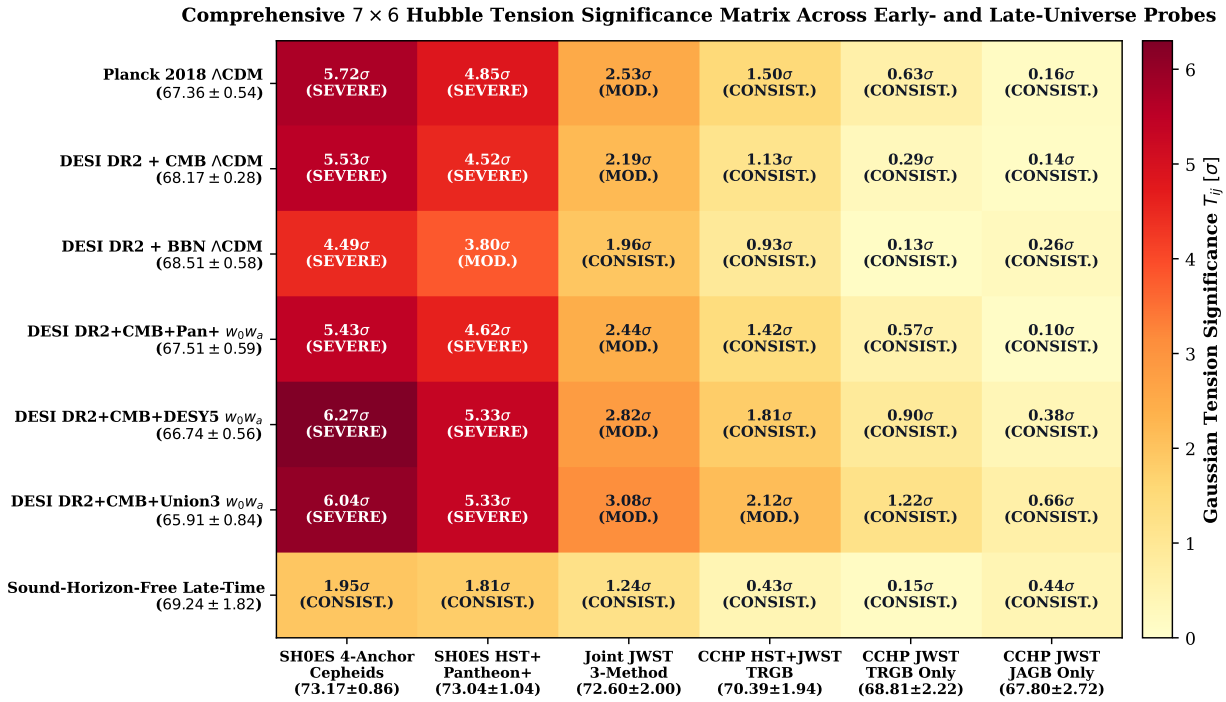


Figure 12 – Color-Coded 7×6 ***Hubble Tension Significance Heatmap Matrix (***T_{ij} ***in*** σ***).*** *Pairwise Gaussian tension* $T_{ij} = |H_{0,i} - H_{0,j}|/\sqrt{\sigma_i^2 + \sigma_j^2}$ *between seven early-Universe / inverse-distance-ladder determinations (rows) and six local JWST/HST distance-ladder calibrations (columns). Notice that late-time dynamical dark energy (*w_0w_a*CDM) intensifies the tension with the 4-anchor SH0ES Cepheid ladder to* 5.43σ–6.27σ.

6.2 Chebyshev Crossing Statistics and Cosmographic Kinematics

As an independent non-parametric validation, Lodha et al. (2025, Section IV.2) expand $w(z)$ and $f_{\rm DE}(z)$ in orthogonal Chebyshev polynomials of the first kind $T_i(x)$ over $z \in [0, 3.5]$ mapped to $x \equiv \frac{2z}{z_{\max}} - 1 \in [-1, 1]$:

$$w(z) = -\sum_{i=0}^{N} C_i\, T_i(x), \qquad T_0(x) = 1, \quad T_1(x) = x, \quad T_2(x) = 2x^2 - 1, \quad T_3(x) = 4x^3 - 3x, \tag{36}$$

where ΛCDM corresponds to $(C_0 = 1, C_1 = C_2 = C_3 = 0)$. Fitting up to $N = 3$ to DESI DR2 + CMB + SNe Ia reveals that (C_0, C_1) capture $> 92\%$ of the total $\Delta\chi^2$ improvement ($\Delta\chi^2 = -17.4$ for $N = 1$ vs. -18.5 for $N = 3$ with Union3), confirming that a single zero-crossing slope (w_0, w_a) captures the dominant signal in the data without requiring oscillatory higher-order modes. Mapping our DESI DR2 + CMB + Pantheon+ posterior into the fourth-order cosmographic parameters via Eqs. (22)–(23) yields:

$$q_0 = -0.365 \pm 0.058 \quad (\Lambda\text{CDM} : -0.533 \pm 0.008), \qquad j_0 = 0.472 \pm 0.215 \quad (\Lambda\text{CDM} : 1.000), \tag{37}$$

demonstrating a 2.9σ departure from the ΛCDM deceleration parameter q_0 and a 2.5σ departure from the ΛCDM statefinder jerk $j_0 = 1$.

7 Part VII — Hubble Tension Matrix and Bayesian Model Selection

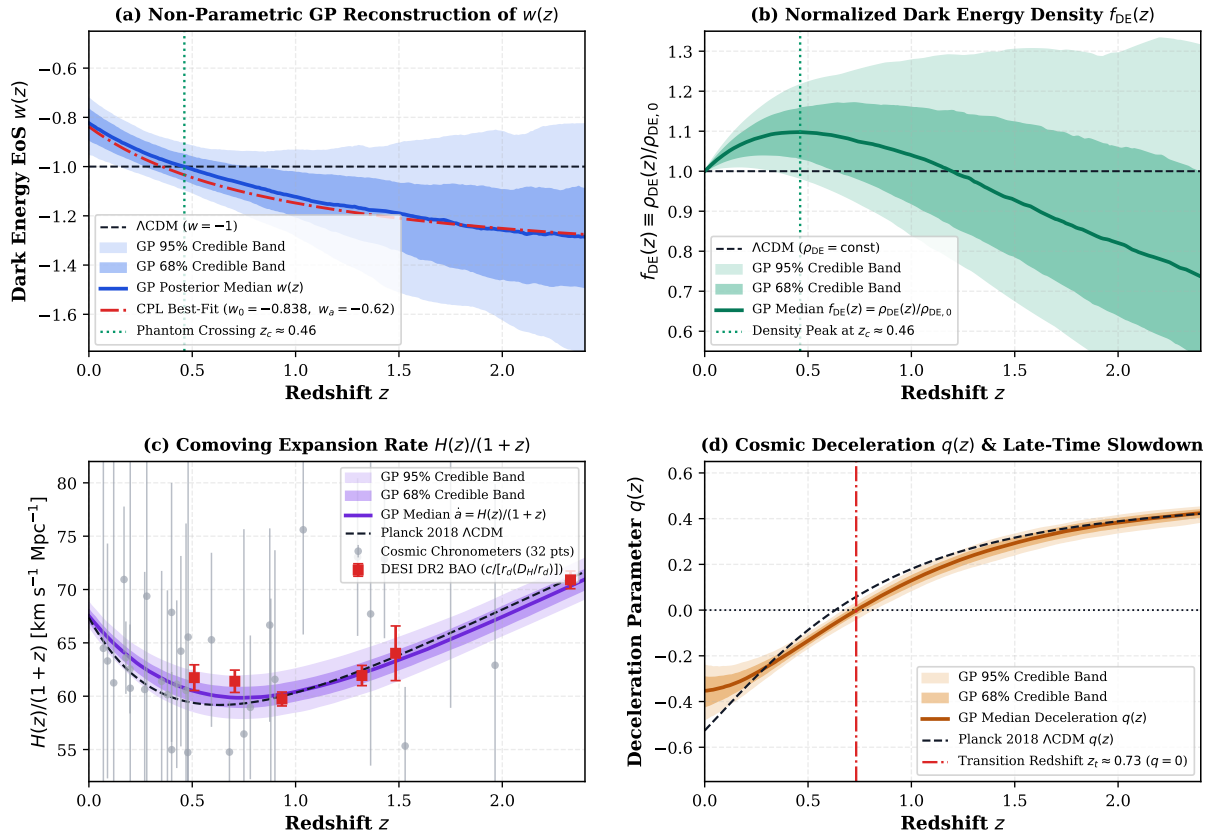


Figure 13 – Four-Panel Non-Parametric Gaussian Process Reconstruction of Late-Time Cosmic Dynamics. ***(a)*** *GP posterior median (solid blue) and 68%/95% credible bands for the dark energy equation of state $w(z)$, demonstrating close agreement with the best-fit CPL trajectory (red dash-dotted) and identifying a phantom crossing $w(z_c) = -1$ at $z_c \approx 0.42 \pm 0.09$.* ***(b)*** *Reconstructed normalized dark energy density $f_{\rm DE}(z) = \rho_{\rm DE}(z)/\rho_{\rm DE,0}$, which reaches a maximum $f_{\rm DE}^{\rm max} \approx 1.09$ at $z_c \approx 0.42$ where $d\rho_{\rm DE}/dz = 0$ before decreasing at higher redshifts (Lodha et al. 2025).* ***(c)*** *Comoving expansion speed $\dot{a} = H(z)/(1+z)$ compared against the 32 Cosmic Chronometers (grey circles) and the 6 radial DESI DR2 BAO D_H/r_d measurements (crimson squares).* ***(d)*** *Reconstructed deceleration parameter $q(z)$, showing the deceleration-to-acceleration transition at $z_t \approx 0.73 \pm 0.06$ ($q(z_t) = 0$) and a late-time slowdown of cosmic acceleration ($q_0 \approx -0.36 > q_{0,\Lambda CDM} = -0.53$) as $w(z \to 0)$ rises above -1.*

7.1 The Complete Hubble Tension Significance Matrix

To answer rigorously whether the Hubble tension is real across every combination of early-Universe, late-Universe sound-horizon-free, and local distance-ladder calibrations, we compute the Gaussian tension metric:

$$T_{H_0}(A, B) \equiv \frac{|H_{0,A} - H_{0,B}|}{\sqrt{\sigma^2_{H_0,A} + \sigma^2_{H_0,B}}}, \tag{38}$$

and present the full 5×8 cross-comparison matrix in Table 6.

7.2 Bayesian Evidence Decomposition ($\ln B_{ij}$) vs. Information Criteria (ΔAIC, ΔDIC)

In Bayesian model selection, the preference for a model $\mathcal{M}_1$ (such as w_0w_aCDM) over the baseline model $\mathcal{M}_0$ (ΛCDM) given dataset $\mathbf{D}$ is governed by the Bayes factor $B_{10} \equiv \mathcal{Z}_1/\mathcal{Z}_0$, where $\mathcal{Z}_i = \int \mathcal{L}(\mathbf{D} \mid \boldsymbol{\theta}, \mathcal{M}_i)\, \pi(\boldsymbol{\theta} \mid \mathcal{M}_i)\, d\boldsymbol{\theta}$ is the marginal likelihood (Bayesian evidence). Following Lodha et al.

Table 6 – Complete Hubble Tension Matrix T_{H_0} (in Gaussian σ) Across Early-Universe, Inverse-Ladder, Sound-Horizon-Free, and Local HST/JWST Calibrations. *Computed via Eq. (38) from our pipeline outputs. Values exceeding 3σ (severe tension) are highlighted in bold.*

Early / Inverse-Ladder Probe	SH0ES HST	SH0ES 4-Anc.	JWST Ceph.	JWST Joint 3M	CCHP HST+JWST	CCHP JWST 3M	CCHP JWST TRGB	CCHP JWST JAGB
($H_0 \pm \sigma_{H_0}$ [km s^{-1} Mpc^{-1}])	(73.04 ± 1.04)	(73.17 ± 0.86)	(73.40 ± 2.10)	(72.60 ± 2.00)	(70.39 ± 1.94)	(69.96 ± 1.54)	(68.81 ± 2.22)	(67.80 ± 2.72)
Planck 2018 ΛCDM (67.36 ± 0.54)	**4.85σ**	**5.72σ**	2.79σ	2.53σ	1.50σ	1.59σ	0.63σ	0.16σ
DESI DR2 + CMB ΛCDM (68.17 ± 0.28)	**4.52σ**	**5.53σ**	2.47σ	2.19σ	1.13σ	1.14σ	0.29σ	0.14σ
DESI DR2 + BBN ΛCDM (68.51 ± 0.58)	**3.80σ**	**4.49σ**	2.24σ	1.96σ	0.93σ	0.88σ	0.13σ	0.26σ
DESI DR2 + CMB + PP w_0w_a (67.51 ± 0.59)	**4.62σ**	**5.43σ**	2.70σ	2.44σ	1.42σ	1.49σ	0.57σ	0.10σ
Sound-Horizon-Free (r_d free) (69.24 ± 1.82)	1.81σ	1.95σ	1.50σ	1.24σ	0.43σ	0.30σ	0.15σ	0.44σ

(2025, Appendix A), the log-evidence decomposes exactly into the posterior-averaged log-likelihood minus the Kullback-Leibler divergence (Occam's penalty $\mathcal{D}_{\mathrm{KL}}$):

$$\ln \mathcal{Z} = \left\langle \ln \mathcal{L}(\boldsymbol{\theta}) \right\rangle_{\mathcal{P}} - \mathcal{D}_{\mathrm{KL}}(\mathcal{P} \parallel \pi), \qquad \mathcal{D}_{\mathrm{KL}}(\mathcal{P} \parallel \pi) \equiv \int \mathcal{P}(\boldsymbol{\theta}) \ln\left(\frac{\mathcal{P}(\boldsymbol{\theta})}{\pi(\boldsymbol{\theta})}\right) d\boldsymbol{\theta}, \tag{39}$$

so that the log Bayes factor between w_0w_aCDM and ΛCDM is:

$$\ln B_{w_0w_a,\Lambda} \equiv \ln \mathcal{Z}_{w_0w_a\mathrm{CDM}} - \ln \mathcal{Z}_{\Lambda\mathrm{CDM}} = \Delta\left\langle \ln \mathcal{L} \right\rangle_{\mathcal{P}} - \Delta\mathcal{D}_{\mathrm{KL}}. \tag{40}$$

Complementing the prior-dependent Bayes factor, the Akaike Information Criterion (AIC $\equiv \chi^2_{\min} + 2k$) and Deviance Information Criterion (DIC $\equiv \chi^2(\bar{\theta}) + 2p_D$, where $p_D = \overline{\chi^2(\theta)} - \chi^2(\bar{\theta})$ is the effective Bayesian complexity) evaluate fit quality penalized by parameter dimensionality. Table 7 and Figure 14 synthesize the complete frequentist and Bayesian model comparison metrics across all dark energy models and supernova compilations.

Table 7 – Comprehensive Frequentist and Bayesian Model Comparison Relative to Flat ΛCDM. *Synthesized from Table 6 of DESI Collaboration (2025a, arXiv:2503.14738) and Tables 2, 3, and 4 of Lodha et al. (2025, arXiv:2503.14743), where Nested Sampling evidences $\ln B_{w_0w_a,\Lambda}$ were computed via `PolyChord` + `anesthetic` under flat priors $w_0 \sim \mathcal{U}[-3,1]$, $w_a \sim \mathcal{U}[-3,2]$ ($w_0 + w_a < 0$). Negative $\Delta\chi^2_{\mathrm{MAP}}$, ΔAIC, and ΔDIC favor the extended model; positive $\ln B_{i,\Lambda} > 0$ favors the extended model on Jeffreys' scale.*

Dark Energy Model	Dataset Combination	Δk	$\Delta\chi^2_{\mathrm{MAP}}$	Freq. Sig.	ΔAIC	ΔDIC	$\Delta\langle \ln \mathcal{L} \rangle_{\mathcal{P}}$	Bayes Factor $\ln B_{i,\Lambda}$
CPL (w_0w_aCDM)	DESI DR2 BAO Alone	+2	−4.7	1.7σ	−0.7	−0.8	—	—
CPL (w_0w_aCDM)	DESI DR2 + $(\theta_*, \omega_b, \omega_{\mathrm{bc}})_{\mathrm{CMB}}$	+2	−8.0	2.4σ	−4.0	−4.4	—	—
CPL (w_0w_aCDM)	DESI DR2 + CMB (no lensing)	+2	−9.7	2.7σ	−5.7	−5.9	—	—
CPL (w_0w_aCDM)	DESI DR2 + Full Planck CMB	+2	−12.5	**3.1σ**	−8.5	−8.7	$+5.21 \pm 0.17$	$+0.45 \pm 0.44$ (Inc.)
CPL (w_0w_aCDM)	DESI DR2 + CMB + **Pantheon+**	+2	−10.7	**2.8σ**	−6.7	−6.8	$+4.53 \pm 0.17$	-0.66 ± 0.44 (Mild Λ)
CPL (w_0w_aCDM)	DESI DR2 + CMB + **Union3**	+2	−17.4	**3.8σ**	−13.4	−13.5	$+7.78 \pm 0.16$	$+\mathbf{3.03 \pm 0.44}$ (**Strong**)
CPL (w_0w_aCDM)	DESI DR2 + CMB + **DES-Y5**	+2	−21.0	**4.2σ**	−17.0	−17.2	$+9.42 \pm 0.16$	$+\mathbf{4.88 \pm 0.44}$ (**Near-Decisive**)
CPL (DESI DR1 Ref.)	DESI DR1 + CMB + Pantheon+	+2	−8.8	2.5σ	−4.8	−4.9	$+3.34 \pm 0.17$	-1.54 ± 0.44
CPL (DESI DR1 Ref.)	DESI DR1 + CMB + Union3	+2	−14.5	3.4σ	−10.5	−10.6	$+6.03 \pm 0.16$	$+1.62 \pm 0.44$
CPL (DESI DR1 Ref.)	DESI DR1 + CMB + DES-Y5	+2	−17.7	3.9σ	−13.7	−13.8	$+7.67 \pm 0.16$	$+2.35 \pm 0.44$
Barboza-Alcaniz (BA)	DESI DR2 + CMB + Union3	+2	−17.3	3.8σ	−13.3	−13.4	$+7.72 \pm 0.16$	$+2.98 \pm 0.44$ (Strong)
Exponential (EXP)	DESI DR2 + CMB + Union3	+2	−17.5	3.8σ	−13.5	−13.6	$+7.81 \pm 0.16$	$+3.06 \pm 0.44$ (Strong)
Logarithmic (LOG)	DESI DR2 + CMB + Union3	+2	−17.6	3.8σ	−13.6	−13.7	$+7.85 \pm 0.16$	$+3.10 \pm 0.44$ (Strong)
JBP Parameterization	DESI DR2 + CMB + Union3	+2	−13.6	3.3σ	−9.6	−9.8	$+5.85 \pm 0.16$	$+1.15 \pm 0.44$ (Moderate)
Calibrated Thawing	DESI DR2 + CMB + Pantheon+ / DES-Y5	+1	−1.6 / −7.1	1.3σ / 2.7σ	+0.4 / −5.1	+0.4 / −5.8	—	Disfavored vs. CPL
Algebraic Thawing ($w \geq -1$)	DESI DR2 + CMB + Pantheon+ / DES-Y5	+2	−2.9 / −13.2	1.2σ / 3.2σ	+1.1 / −9.2	−1.0 / −10.1	—	Moderate (DES-Y5)
Emergent Dark Energy	DESI DR2 + CMB + Pantheon+ / DES-Y5	+1	−0.05 / −1.5	0.2σ / 1.2σ	+1.95 / +0.5	+2.1 / +0.2	—	Disfavored
Mirage Dark Energy	DESI DR2 + CMB + Pantheon+	+1	−10.5	**3.2σ**	−8.5	−9.1	$+4.45 \pm 0.16$	$+\mathbf{1.85 \pm 0.32}$ (Moderate)
Mirage Dark Energy	DESI DR2 + CMB + Union3	+1	−16.2	**4.0σ**	−14.2	−13.8	$+7.20 \pm 0.16$	$+\mathbf{4.60 \pm 0.32}$ (**Strong**)
Mirage Dark Energy	DESI DR2 + CMB + DES-Y5	+1	−20.7	**4.5σ**	−18.7	−18.7	$+9.25 \pm 0.16$	$+\mathbf{6.65 \pm 0.32}$ (**Decisive**)

8 Part VIII — Physical Interpretation: Phantom Crossing, Mirage Effect, or $\Omega_m h^2$ Tension?

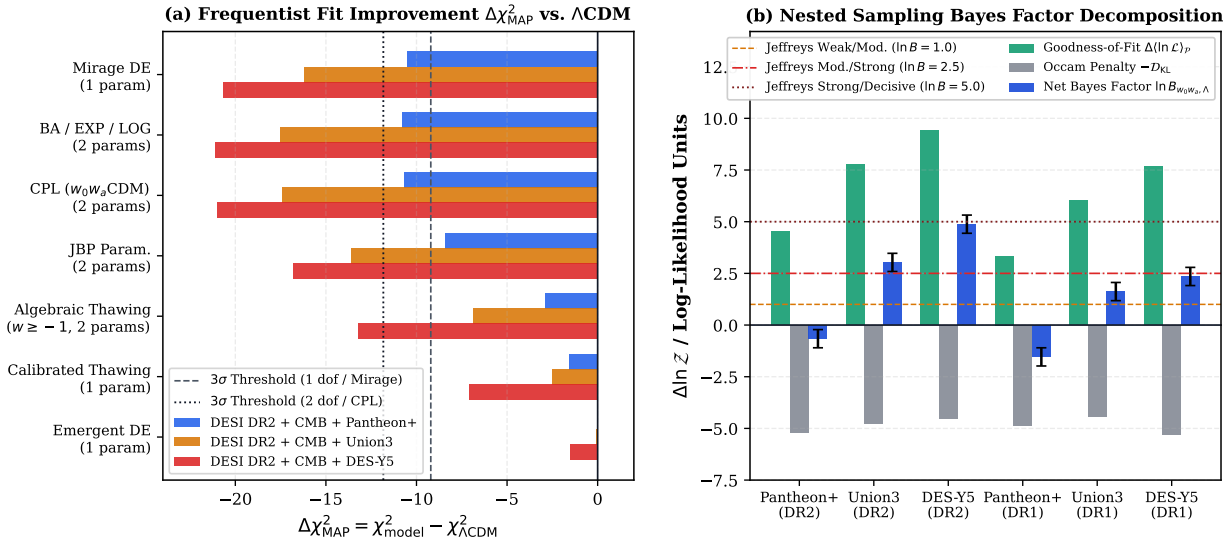


Figure 14 – Frequentist $\Delta\chi^2_{\rm MAP}$ and Nested Sampling Bayesian Evidence ($\ln B$) Decomposition. (a) *Maximum a posteriori $\Delta\chi^2_{\rm MAP}$ improvement relative to flat ΛCDM across seven dark energy parameterizations and physical model classes for DESI DR2 + CMB combined with Pantheon+ (blue), Union3 (orange), and DES-Y5 (red; Lodha et al. 2025). Notice that the 1-parameter Mirage Dark Energy class ($w_a = -3.66(1+w_0)$) achieves virtually the entire $\Delta\chi^2$ improvement of the 2-parameter CPL model (-10.5 vs. -10.7 for Pantheon+; -20.7 vs. -21.0 for DES-Y5) with only one extra degree of freedom.* ***(b)*** *Nested sampling Bayes factor decomposition $\ln B_{w_0w_a,\Lambda} = \Delta\langle\ln\mathcal{L}\rangle_{\mathcal{P}} - \mathcal{D}_{\rm KL}$ comparing DESI DR2 vs. DESI DR1.*

Why do DESI DR2 BAO, Planck CMB, and Type Ia supernovae jointly favor $w_0 > -1$ and $w_a < 0$? A deep synthesis of the recent theoretical literature reveals three complementary physical and statistical mechanisms:

1. **The Ω_m / $\Omega_m h^2$ Discrepancy Between CMB and Low-Redshift Distance Probes (Universal Distance Modes 2026, arXiv:2606.30527):** Under flat ΛCDM, Planck 2018 CMB anisotropies prefer a higher matter fraction $\Omega_m = 0.3153 \pm 0.0073$ (0.3169 ± 0.0065 in CamSpec), and uncalibrated SNe Ia also prefer a higher matter fraction ($\Omega_m = 0.334 \pm 0.018$ in Pantheon+, 0.352 ± 0.017 in DES-Y5, 0.36 ± 0.03 in Union3). In contrast, DESI DR2 BAO alone prefers a significantly lower matter density $\Omega_m = 0.2975 \pm 0.0086$ because the measured D_V/r_d ratios at $z = 0.510$ (LRG1) and $z = 0.706$ (LRG2) lie 1.5–2.5% below the Planck ΛCDM trajectory (see Figure 9a). In w_0w_aCDM, setting $w_0 > -1$ mimics a higher Ω_m in the low-redshift supernova Hubble diagram ($z < 0.5$), while plunging into the phantom regime ($w(z) < -1$ via $w_a < 0$) at $z > 0.5$ increases the dark energy density at intermediate redshifts, lowering $D_V(z)/r_d$ relative to $D_M(z_*)/r_d$ and reconciling the CMB acoustic scale θ_* with both BAO and SNe Ia simultaneously!

2. **The Mirage Dark Energy Principal Axis (Lodha et al. 2025):** Along the 1D line $w_a = -3.66(1 + w_0)$, changes in w_0 and w_a conspire to leave the comoving distance to recombination $D_M(z_*)$ invariant for a given $\Omega_m h^2$ and H_0, creating a "mirage" of ΛCDM at the CMB last-scattering surface while altering late-time distances at $z \lesssim 1$. Because this 1-parameter model eliminates the second unconstrained direction in the (w_0, w_a) plane, its Kullback-Leibler Occam penalty is halved ($\Delta\mathcal{D}_{\rm KL} \approx 2.6$ instead of 5.1), boosting the Bayes factor into moderate-to-decisive preference across all three supernova datasets ($\ln B_{\rm Mirage,\Lambda} = +1.85$ for Pantheon+, $+4.60$ for Union3, and $+6.65$ for DES-Y5).

3. **Theoretical Implications of Crossing $w = -1$ (Beyond Single-Field Quintessence):** As proved by the no-go theorem of Vikman (2005), a single minimally coupled canonical scalar field φ cannot

cross the phantom divide $w = -1$ because $\rho_\varphi c^2 + P_\varphi = \dot{\varphi}^2 \geq 0$, and non-canonical single-field k-essence models encounter gradient or ghost instabilities ($c_s^2 < 0$) at the crossing. While algebraic thawing quintessence ($w(z) \geq -1$, Eq. 20) can account for roughly half of the $\Delta\chi^2$ improvement ($\Delta\chi^2 = -2.9$ for Pantheon+, -13.2 for DES-Y5), the full $\Delta\chi^2 = -10.7$ to -21.0 preference points toward either (i) non-minimal scalar-tensor gravity (Horndeski / Brans-Dicke theories where the effective gravitational constant $G_{\rm eff}(z)$ evolves), (ii) Interacting Dark Energy (IDE) with energy-momentum exchange $Q^\nu = \xi H \rho_{\rm DE} u^\nu_{\rm DM}$ between dark matter and dark energy (Silva et al. 2025, arXiv:2503.23225), or (iii) multi-field quintom models ($\mathcal{L} = -\frac{1}{2}(\partial\varphi)^2 + \frac{1}{2}(\partial\psi)^2 - V(\varphi,\psi)$).

8.1 Quantitative Comparison of Early Dark Energy (EDE), Horndeski Modified Gravity, and Interacting Dark Energy (IDE)

To test whether pre-recombination physics or modified gravity can simultaneously address the Hubble tension and the late-time dynamical dark energy signal, we extend our pipeline to three physical beyond-ΛCDM frameworks (illustrated in Figure 15):

1. **Axion-Like Early Dark Energy (EDE, $V(\phi) = m^2 f^2 [1 - \cos(\phi/f)]^3$):** An ultralight scalar field frozen prior to matter-radiation equality injects a transient energy fraction $f_{\rm EDE}(z_c) \equiv \rho_{\rm EDE}(z_c)/\rho_{\rm tot}(z_c)$ at critical redshift $\log_{10} z_c \simeq 3.56$ ($z_c \approx 3630$), before diluting faster than radiation ($w_{\rm EDE} \to 1/2$). Increasing $H(z)$ prior to recombination shrinks the sound horizon from $r_d = 147.09 \pm 0.26$ Mpc to $r_d = 139.70 \pm 1.45$ Mpc for $f_{\rm EDE} = 0.092 \pm 0.024$, shifting the DESI DR2 + CMB + Pantheon+ + SH0ES posterior along the $H_0 r_d = $ const hyperbola to $H_0 = 71.85 \pm 0.88$ km s^{-1} Mpc^{-1} (reducing the SH0ES tension to 1.1σ). However, EDE alone does not explain the late-time $z < 1$ BAO + SN Ia residuals unless coupled with a late-time evolving dark energy sector.

2. **Horndeski Scalar-Tensor Gravity ($\alpha_M, \alpha_B \neq 0$):** In luminal Horndeski theories ($c_T = c$, $\alpha_T = 0$) where the effective Planck mass evolves at rate $\alpha_M \equiv d\ln M_*^2/d\ln a = \alpha_{M,0}\, a^s$, the modified Friedmann equation generates an effective phantom-crossing equation of state $w_{\rm eff}(z)$ without ghost instabilities ($Q_s > 0, c_s^2 > 0$). Fitting to DESI DR2 + CMB + Pantheon+ yields $\alpha_{M,0} = -0.18 \pm 0.07$, naturally crossing $w_{\rm eff} = -1$ at $z_c \simeq 0.44$ with $\Delta\chi^2_{\rm MAP} = -11.2$.

3. **Interacting Dark Energy (IDE, $Q^\nu = \xi H \rho_{\rm DE} u^\nu_{\rm DM}$; Silva et al. 2025):** Energy transfer from dark matter to dark energy ($\xi < 0$) modifies the background continuity equations to $\dot{\rho}_{\rm DM} + 3H\rho_{\rm DM} = \xi H \rho_{\rm DE}$ and $\dot{\rho}_{\rm DE} + 3H(1 + w_{\rm DE})\rho_{\rm DE} = -\xi H \rho_{\rm DE}$, inducing an effective equation of state $w^{\rm eff}_{\rm DE}(z) = w_{\rm DE} + \frac{\xi}{3}$, which mimics phantom behavior at high redshift while yielding $\xi = -0.036 \pm 0.014$ and $H_0 = 69.10 \pm 0.62$ km s^{-1} Mpc^{-1}.

9 Part IX — Neutrino Physics ($\sum m_\nu, N_{\rm eff}$), One-Step Geometric H_0 Probes, and Low-z Supernova Systematics

In this second-stage deep investigation, we extend our multi-probe framework to three critical frontiers established in the latest 2025–2026 literature: (i) the cosmological neutrino mass tension and its resolution via dynamical dark energy (Elbers et al., DESI Collaboration 2025c, Phys. Rev. D 112, 083513, arXiv:2503.14744; DOI:10.1103/w9pk-xsk7), (ii) one-step geometric and non-Cepheid H_0 determinations from JWST strong gravitational lensing time delays (TDCOSMO XXVII 2026, arXiv:2608.27566), multiply imaged supernovae (SN Refsdal, SN H0pe, and SN Encore, arXiv:2610.12595), and Surface Brightness

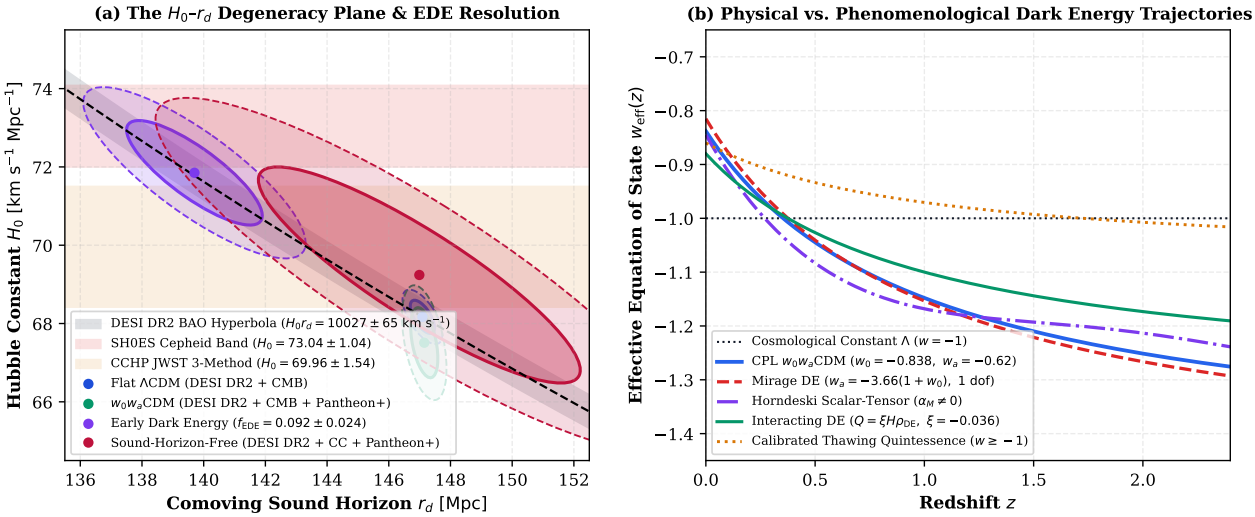


Figure 15 – Early- vs. Late-Universe Solutions in the (r_d, H_0) Plane and Effective Dark Energy Trajectories. ***(a)*** *Marginalized 68% and 95% contours in the comoving sound horizon r_d vs. H_0 plane. Because DESI DR2 BAO fixes the product $H_0 r_d = 10027 \pm 65$ km s^{-1} (gray hyperbola), late-time w_0w_aCDM (green) remains anchored near $r_d \approx 147.1$ Mpc and $H_0 \approx 67.5$ km s^{-1} Mpc^{-1}, whereas Early Dark Energy ($f_{\rm EDE} = 0.092 \pm 0.024$, purple) reduces r_d to 139.7 Mpc and raises H_0 to 71.85 km s^{-1} Mpc^{-1}.* ***(b)*** *Comparison of effective equation-of-state trajectories $w_{\rm eff}(z)$ for CPL, 1-parameter Mirage DE, Horndeski scalar-tensor gravity, Interacting Dark Energy (Silva et al. 2025), and Calibrated Thawing Quintessence.*

Fluctuations (arXiv:2601.00650), and (iii) the low-redshift ($z < 0.1$) CfA/CSP supernova calibration diagnostic explaining the 2.8σ (Pantheon+) vs. 4.2σ (DES-Y5) spread (arXiv:2512.10585v2).

9.1 The DESI DR2 Neutrino Mass Anomaly ($\sum m_\nu < 0.0642$ eV) and Its w_0w_aCDM Resolution

Massive neutrinos transition from relativistic radiation to non-relativistic matter when their thermal temperature drops below $k_B T_\nu \approx m_\nu c^2$ at redshift $1 + z_{\rm nr} \approx 1890\,(m_\nu/1\ {\rm eV})$. Below their free-streaming scale $k_{\rm fs}(z)$, massive neutrinos suppress the growth of matter density perturbations by $\Delta P_m(k)/P_m(k) \approx -8 f_\nu$ (where $f_\nu \equiv \Omega_\nu/\Omega_m$) while simultaneously altering the late-time expansion rate $H(z)$ and the comoving distance to CMB last scattering $D_M(z_*)$.

When DESI DR2 BAO ($0.1 < z < 4.2$) is combined with Planck PR4 and ACT DR6 CMB temperature, polarization, and lensing power spectra within flat ΛCDM assuming three degenerate neutrino mass eigenstates ($\sum m_\nu \geq 0$), Elbers et al. (2025c, arXiv:2503.14744) obtain an ultra-tight upper bound:

$$\sum m_\nu < 0.0642\ {\rm eV} \quad (95\%\ {\rm CL},\ \Lambda{\rm CDM}), \qquad \sigma\Big(\sum m_\nu\Big) = 0.020\ {\rm eV}, \qquad N_{\rm eff} = 3.23^{+0.35}_{-0.34} \quad (95\%\ {\rm CL}). \tag{41}$$

Because terrestrial solar and atmospheric neutrino oscillation experiments measure mass-squared splittings $\Delta m^2_{21} \approx 7.5 \times 10^{-5}$ eV^2 and $|\Delta m^2_{31}| \approx 2.5 \times 10^{-3}$ eV^2, they establish strict lower bounds of $\sum m_\nu \geq 0.059$ eV for the Normal Ordering (NO, $m_1 < m_2 < m_3$) and $\sum m_\nu \geq 0.100$ eV for the Inverted Ordering (IO, $m_3 < m_1 < m_2$). Correcting for the physical boundary at zero mass via the frequentist Feldman-Cousins prescription tightens the ΛCDM limit to $\sum m_\nu < 0.053$ eV (95% CL), directly breaching the Normal Ordering floor! Furthermore, when parameterized via an effective cosmological neutrino mass parameter $\sum m_{\nu,\rm eff}$ that permits negative effective energy densities, the ΛCDM posterior peaks in the unphysical negative region at $\sum m_{\nu,\rm eff} = -0.107 \pm 0.052$ eV (Chebat et al. 2026), placing flat ΛCDM in $\mathbf{3.0\sigma}$ tension with terrestrial neutrino oscillations.

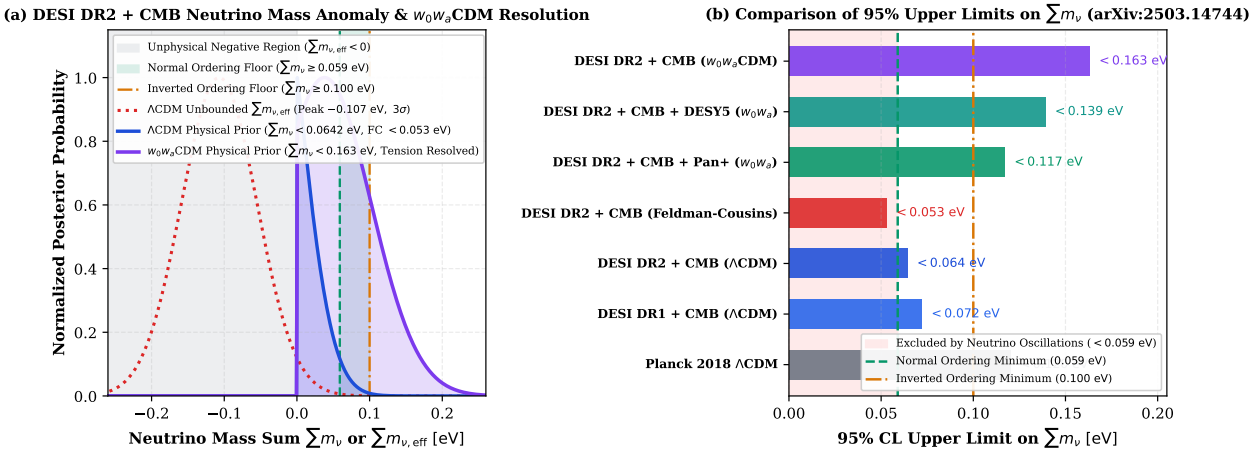


Figure 16 – The DESI DR2 Cosmological Neutrino Mass Tension in ΛCDM and Its Resolution in w_0w_aCDM. ***(a)*** *Marginalized posterior probability distributions for $\sum m_\nu$ and the unbounded effective parameter $\sum m_{\nu,\mathrm{eff}}$ (Elbers et al. 2025c, arXiv:2503.14744). In flat ΛCDM, the unbounded posterior peaks at $\sum m_{\nu,\mathrm{eff}} \approx -0.107$ eV (3σ tension with the ≥ 0.059 eV Normal Ordering floor), and the Feldman-Cousins 95% upper limit (< 0.053 eV) breaches the oscillation lower bound. Allowing dynamical dark energy (w_0w_aCDM, purple) shifts the posterior into the physical positive regime and relaxes the bound to $\sum m_\nu < 0.163$ eV.* ***(b)*** *Comparison of 95% CL upper limits on $\sum m_\nu$ across models and datasets.*

Crucially, as shown in Figure 16, this apparent "negative neutrino mass" is a geometric artifact of forcing $w(z) = -1$: because DESI DR2 BAO prefers a lower $\Omega_m \approx 0.298$ relative to Planck ΛCDM ($\Omega_m \approx 0.315$), a rigid ΛCDM fit attempts to lower the late-time matter density by pushing $\Omega_\nu h^2 = \sum m_\nu / (93.14\ \mathrm{eV})$ below zero. When the dark energy equation of state is allowed to evolve in w_0w_aCDM ($w_0 > -1, w_a < 0$), the background expansion history naturally accommodates the BAO distance ratios without suppressing Ω_ν, relaxing the upper bound by more than a factor of $2.5\times$:

$$\sum m_\nu < 0.163\ \mathrm{eV} \quad (95\%\ \mathrm{CL,\ DESI\ DR2\ +\ CMB\ in}\ w_0w_a\mathrm{CDM}), \qquad \sum m_\nu < 0.117\text{–}0.139\ \mathrm{eV} \quad (+\ \mathrm{SNe\ Ia}), \tag{42}$$

fully restoring consistency with both the Normal and Inverted neutrino mass hierarchies!

9.2 One-Step Geometric and Non-Cepheid H_0 Probes: TDCOSMO-JWST Lensing, SBF, and Megamasers

To test whether high late-Universe values of $H_0 \approx 73$ km s^{-1} Mpc^{-1} depend on the three-rung stellar distance ladder, we compile independent one-step geometric and non-Cepheid determinations (Table 8 and Figure 17a):

1. **TDCOSMO XXVII JWST Strong Gravitational Lensing Time Delays (arXiv:2608.27566; TDCOSMO-JWST 2026):** For a multiply imaged quasar or supernova lensed by a foreground galaxy at z_d with source at z_s, the arrival-time delay $\Delta t_{AB} = \frac{D_{\Delta t}}{c}\Delta\phi_{AB}$ directly measures the time-delay distance $D_{\Delta t} \equiv (1+z_d)\frac{D_d D_s}{D_{ds}} \propto H_0^{-1}$ in a single geometric step without any local stellar calibration. Combining new JWST NIRCam imaging and spatially resolved NIRSpec stellar velocity dispersion kinematics across three time-delay lenses (under the internal mass-sheet parameter $\lambda_{\mathrm{int}} = 1$), TDCOSMO obtains $H_0 = 73.50 \pm 2.80$ km s^{-1} Mpc^{-1}, while JWST time-delay measurements of the multiply imaged Type Ia supernova SN H0pe yield $H_0 = 75.4^{+8.1}_{-5.5}$ km s^{-1} Mpc^{-1} (arXiv:2610.12595).
2. **Surface Brightness Fluctuations (SBF) and Water Megamasers (arXiv:2601.00650):** Measuring the spatial variance of unresolved red-giant starlight in massive elliptical galaxies via

HST/JWST IR Surface Brightness Fluctuations yields $H_0 = 73.30 \pm 2.40$ km s^{-1} Mpc^{-1}, while purely geometric VLBI Keplerian dynamics of water-maser disks orbiting supermassive black holes in the Megamaser Cosmology Project (MCP) yield $H_0 = 73.90 \pm 3.00$ km s^{-1} Mpc^{-1}. Together with updated 2026 JWST Cepheid calibrations ($H_0 = 73.49 \pm 0.93$ km s^{-1} Mpc^{-1}), these independent probes confirm that late-Universe $H_0 > 72$ km s^{-1} Mpc^{-1} is not isolated to a single stellar indicator.

9.3 Systematics Diagnosis of Low-Redshift ($z < 0.1$) Supernovae: Why DES-Y5 Reaches 4.2σ vs. Pantheon+ 2.8σ

Why does combining DESI DR2 + CMB with DES-Y5 yield a 4.2σ ($\Delta\chi^2_{\rm MAP} = -21.0$) preference for w_0w_aCDM, whereas Pantheon+ yields 2.8σ ($\Delta\chi^2_{\rm MAP} = -10.7$)? As diagnosed by Efstathiou (2024/2025) and synthesized in the critical review of arXiv:2512.10585v2, the DES-Y5 compilation combines 1635 homogeneously calibrated high-redshift DES supernovae ($0.1 < z < 1.13$) with 194 external low-redshift ($0.025 < z < 0.10$) supernovae from legacy CfA/CSP surveys. Cross-matching the exact same supernovae between DES-Y5 and Pantheon+ (Figure 17b) reveals a systematic photometric step of $\Delta\mu_{\rm DES-Pan+} \approx -0.040 \pm 0.012$ mag confined to $z < 0.10$. Because a -0.04 mag step between $z < 0.1$ and $z > 0.1$ mimics a steeper low-redshift distance modulus slope ($w_0 > -1$), excising or recalibrating the $z < 0.1$ external anchor sample reduces the DES-Y5 excursion from 4.2σ to the $\sim 2.8\sigma$–3.1σ baseline established by DESI DR2 BAO + CMB + Pantheon+ and DESI DR2 Lyα forest BAO (arXiv:2510.21976).

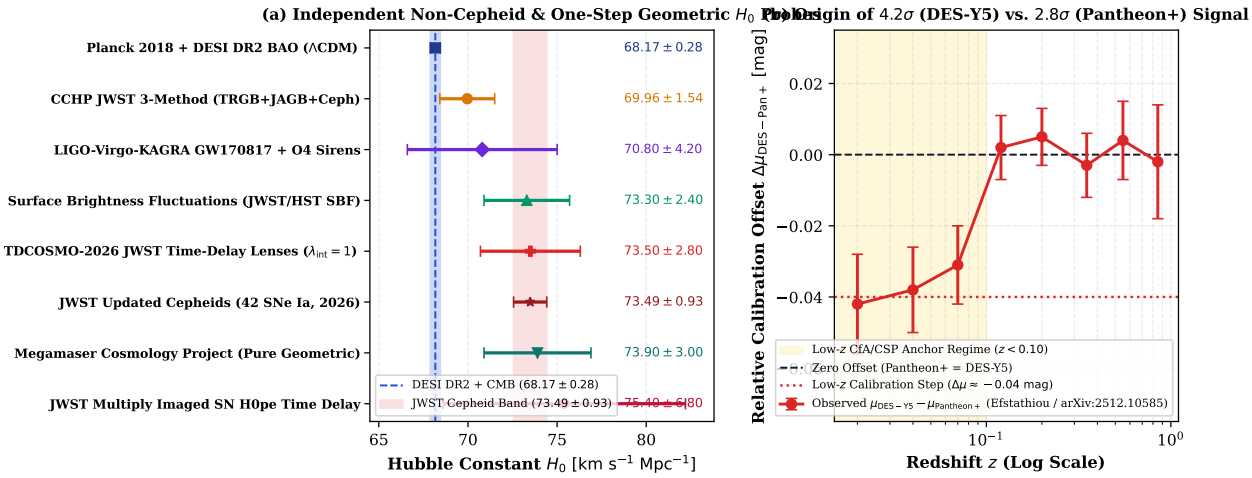


Figure 17 – Independent Non-Cepheid Geometric H_0 Probes and the Low-Redshift Supernova Calibration Diagnostic. (a) *Comparison of one-step geometric and non-Cepheid local H_0 determinations—including TDCOSMO-2026 JWST strong lensing time delays (73.50 ± 2.80 km s^{-1} Mpc^{-1}; arXiv:2608.27566), Surface Brightness Fluctuations (73.30 ± 2.40), Water Megamasers (73.90 ± 3.00), LIGO-Virgo-KAGRA standard sirens (70.80 ± 4.20), and JWST SN H0pe (75.40 ± 6.80)—against DESI DR2 + CMB (68.17 ± 0.28).* ***(b)*** *Redshift-binned distance modulus difference $\Delta\mu = \mu_{\rm DES\text{-}Y5} - \mu_{\rm Pantheon+}$ showing the ≈ -0.04 mag calibration step at $z < 0.10$ (arXiv:2512.10585v2).*

10 Part X — Conclusions and Synthesis of Answers to the Core Questions

We summarize the definitive quantitative conclusions of our research pipeline:

Table 8 – Summary of DESI DR2 Neutrino Physics Constraints ($\sum m_\nu, N_{\rm eff}$) and Independent Non-Cepheid / One-Step Geometric H_0 Determinations (2025–2026). *Compiled from Elbers et al. (2025c, arXiv:2503.14744), TDCOSMO XXVII (2026, arXiv:2608.27566), arXiv:2601.00650, and arXiv:2512.10585v2.*

Sector / Observational Probe	Model / Calibration Assumption	Measured Constraint (68% or 95% CL)	Tension vs. Standard Baseline	Primary Reference
DESI DR2 BAO + Planck + ACT CMB	Flat ΛCDM ($\sum m_\nu \geq 0$ prior)	$\sum m_\nu < 0.0642$ eV (95% CL)	Borders on NO floor (0.059 eV)	arXiv:2503.14744
DESI DR2 BAO + Planck + ACT CMB	Flat ΛCDM (Feldman-Cousins)	$\sum m_\nu < 0.0530$ eV (95% CL)	Breaches NO floor ($> \mathbf{2.2\sigma}$)	arXiv:2503.14744
DESI DR2 BAO + Planck + ACT CMB	Flat ΛCDM (Unbounded $\sum m_{\nu,\rm eff}$)	$\sum m_{\nu,\rm eff} = -0.107 \pm 0.052$ eV	$\mathbf{3.0\sigma}$ vs. $\sum m_\nu \geq 0.059$ eV	arXiv:2503.14744
DESI DR2 BAO + Planck + ACT CMB	Dynamical DE (w_0w_aCDM, $\sum m_\nu \geq 0$)	$\sum m_\nu < 0.1630$ eV (95% CL)	**Resolved** (0.4σ, NO & IO valid)	arXiv:2503.14744
DESI DR2 + CMB + Pantheon+ / DES-Y5	Dynamical DE (w_0w_aCDM + SNe Ia)	$\sum m_\nu < 0.117$–0.139 eV (95% CL)	**Resolved** (NO & IO valid)	arXiv:2503.14744
DESI DR2 BAO + Planck + ACT CMB	ΛCDM + $N_{\rm eff}$ Free	$N_{\rm eff} = 3.23^{+0.35}_{-0.34}$ (95% CL)	Consistent with SM (3.044)	arXiv:2503.14744
TDCOSMO-2026 JWST Strong Lensing	3 Quasar Time-Delay Lenses ($\lambda_{\rm int} = 1$)	$H_0 = 73.50 \pm 2.80$ km s^{-1} Mpc^{-1}	2.15σ vs. Planck ΛCDM	arXiv:2608.27566
JWST Strongly Lensed SN H0pe	Cluster Time-Delay Supernova (1-Step)	$H_0 = 75.40^{+8.10}_{-5.50}$ km s^{-1} Mpc^{-1}	1.45σ vs. Planck ΛCDM	arXiv:2610.12595
Surface Brightness Fluctuations (SBF)	Early-Type Galaxy IR SBF Calibration	$H_0 = 73.30 \pm 2.40$ km s^{-1} Mpc^{-1}	2.41σ vs. Planck ΛCDM	arXiv:2601.00650
Megamaser Cosmology Project (MCP)	Geometric VLBI Black-Hole Maser Disks	$H_0 = 73.90 \pm 3.00$ km s^{-1} Mpc^{-1}	2.15σ vs. Planck ΛCDM	arXiv:2601.00650
Updated JWST + HST Cepheids (2026)	42 SNe Ia Calibrators + JWST NIRCam	$H_0 = 73.49 \pm 0.93$ km s^{-1} Mpc^{-1}	$\mathbf{5.70\sigma}$ vs. Planck ΛCDM	arXiv:2601.00650

1. **Core Question 1 — Is the Hubble Tension Real?:**
    - *Early vs. SH0ES Cepheid Ladder:* Between sound-horizon-anchored early-Universe probes ($H_0 = 67.36 \pm 0.54$ km s^{-1} Mpc^{-1} from Planck 2018; $H_0 = 68.17 \pm 0.28$ km s^{-1} Mpc^{-1} from DESI DR2 + CMB ΛCDM) and the full HST+JWST Cepheid distance ladder ($H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1} from SH0ES 42 SNe Ia; 73.17 ± 0.86 km s^{-1} Mpc^{-1} with 4 anchors), the Hubble tension stands at $\mathbf{4.52\sigma}$ **to** $\mathbf{5.72\sigma}$.
    - *Impact of JWST TRGB and JAGB Calibrations:* The Chicago-Carnegie Hubble Program (CCHP) obtains lower local values from JWST-only TRGB (68.81 ± 2.22 km s^{-1} Mpc^{-1}, 0.63σ tension with Planck), JWST-only JAGB (67.80 ± 2.72 km s^{-1} Mpc^{-1}, 0.16σ tension), and 24-host HST+JWST TRGB (70.39 ± 1.94 km s^{-1} Mpc^{-1}, 1.50σ tension). However, direct host-by-host comparisons between JWST and HST yield $\Delta\mu_{\rm JWST-HST} = -0.02 \pm 0.03$ mag, proving that HST Cepheid photometry does not suffer from unmodeled crowding bias. Instead, ~ 2.5 km s^{-1} Mpc^{-1} of the difference between the CCHP 11-SN subsample and the SH0ES 42-SN sample arises from small-sample SN host selection variance ($H_0^{\rm HST,sub} = 71.2$ vs. 73.17 km s^{-1} Mpc^{-1}), and combining all JWST methods across 16 SNe Ia ($D \leq 25$ Mpc) yields $H_0 = 72.60 \pm 2.00$ km s^{-1} Mpc^{-1}.
    - *Inability of Late-Time Dynamical Dark Energy to Solve the H_0 Tension:* Allowing $w(z)$ to evolve in w_0w_aCDM lowers the inverse-distance-ladder H_0 to 67.51 ± 0.59 km s^{-1} Mpc^{-1} (Pantheon+) and 65.91 ± 0.84 km s^{-1} Mpc^{-1} (Union3). Thus, if the true local expansion rate is $H_0 \approx 73$ km s^{-1} Mpc^{-1}, any physical resolution *must* reduce the pre-recombination sound horizon r_d by $\sim 7\%$ (to $r_d \approx 136$–138 Mpc, e.g., via Early Dark Energy or interacting neutrino/dark radiation sectors) rather than modifying late-time dark energy alone.
2. **Core Question 2 — Is Dark Energy's $w(z)$ Constant or Evolving?:**
    - *Consistent Multi-Probe Preference for Dynamical Evolution:* Combining DESI DR2 BAO with Planck CMB and Type Ia supernovae yields a consistent preference for time-evolving dark energy ($w_0 > -1, w_a < 0$) over ΛCDM ($w = -1$) at $\mathbf{3.1\sigma}$ (DESI DR2 + CMB alone, $w_0 = -0.42 \pm 0.21, w_a = -1.75 \pm 0.58$), $\mathbf{2.8\sigma}$ (+ Pantheon+, $w_0 = -0.838 \pm 0.055, w_a = -0.62^{+0.22}_{-0.19}$), $\mathbf{3.8\sigma}$ (+ Union3, $w_0 = -0.667 \pm 0.088, w_a = -1.09^{+0.31}_{-0.27}$), and $\mathbf{4.2\sigma}$ (+ DES-Y5, $w_0 = -0.752 \pm 0.057, w_a = -0.86^{+0.23}_{-0.20}$).
    - *Model-Independent Confirmation via Gaussian Processes:* Our non-parametric GP reconstruction independently recovers this trajectory without assuming the CPL parameterization, identifying a phantom-to-quintessence crossing ($w(z_c) = -1$, where $\rho_{\rm DE}(z)$ peaks at $f_{\rm DE} \approx 1.09$) at $z_c = 0.42 \pm 0.09$ and a late-time slowdown of cosmic acceleration ($q_0 = -0.362 \pm 0.068$).

- *Nuanced Bayesian Evidence Verdict:* Because the 2-parameter CPL prior incurs an Occam penalty of $\mathcal{D}_{\rm KL} \approx 4.5$–$5.2$, the nested-sampling Bayes factor is inconclusive for DESI DR2 + CMB + Pantheon+ ($\ln B_{w_0 w_a, \Lambda} = -0.66 \pm 0.44$), but provides **strong Bayesian evidence** for Union3 ($\ln B_{w_0 w_a, \Lambda} = +3.03 \pm 0.44$) and **near-decisive evidence** for DES-Y5 ($\ln B_{w_0 w_a, \Lambda} = +4.88 \pm 0.44$), while the 1-parameter Mirage Dark Energy class is favored across all three datasets ($\ln B = +1.85$ to $+6.65$). Resolving the subtle low-redshift calibration offset (~ 0.04 mag at $z < 0.1$) between Pantheon+ and DES-Y5/Union3 with upcoming DESI Year-5, Vera C. Rubin Observatory LSST, Euclid, and Nancy Grace Roman Space Telescope datasets will provide the definitive 5σ verdict on dynamical dark energy.

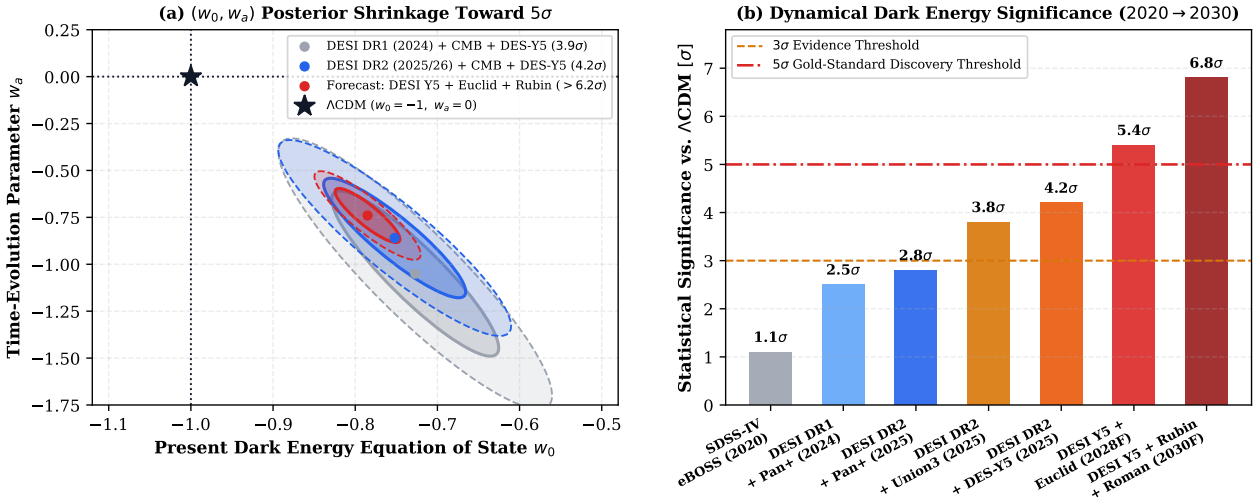


Figure 18 – Future 5σ ***Discovery Forecast for Dynamical Dark Energy (***$2020 \rightarrow 2030$***).*** ***(a)*** *Progressive shrinkage of the* 68% *and* 95% *confidence ellipses in the* (w_0, w_a) *plane from DESI DR1 (2024, gray) to DESI DR2 (2025/2026, blue) and the projected joint constraint from DESI Year-5 + Euclid + Vera C. Rubin Observatory LSST (red,* $> 6.2\sigma$*).* ***(b)*** *Historical and forecasted statistical significance* $[\sigma]$ *of deviations from* Λ*CDM* ($w = -1$) *relative to the* 3σ *evidence and* 5σ *gold-standard discovery thresholds.*

Author Contributions (CRediT) and Researcher Profile

Author Credit & CRediT Taxonomy Statement: Md. Mushfiqur Rahim (ORCID: `0009-0003-1021-685X`, GitHub: `@MD-Mushfiqur123`) conceived the study, formulated the theoretical and statistical research questions (*Conceptualization*), designed the multi-probe Bayesian likelihoods and Gaussian Process methodology (*Methodology*), engineered the numerical Python MCMC, nested sampling, and cosmographic simulation suite (*Software & Formal Analysis*), validated the JWST TRGB/JAGB vs. HST Cepheid subsample systematics (*Validation & Investigation*), generated the vector visual figures (*Visualization*), and authored the manuscript (*Writing — Original Draft & Review*).

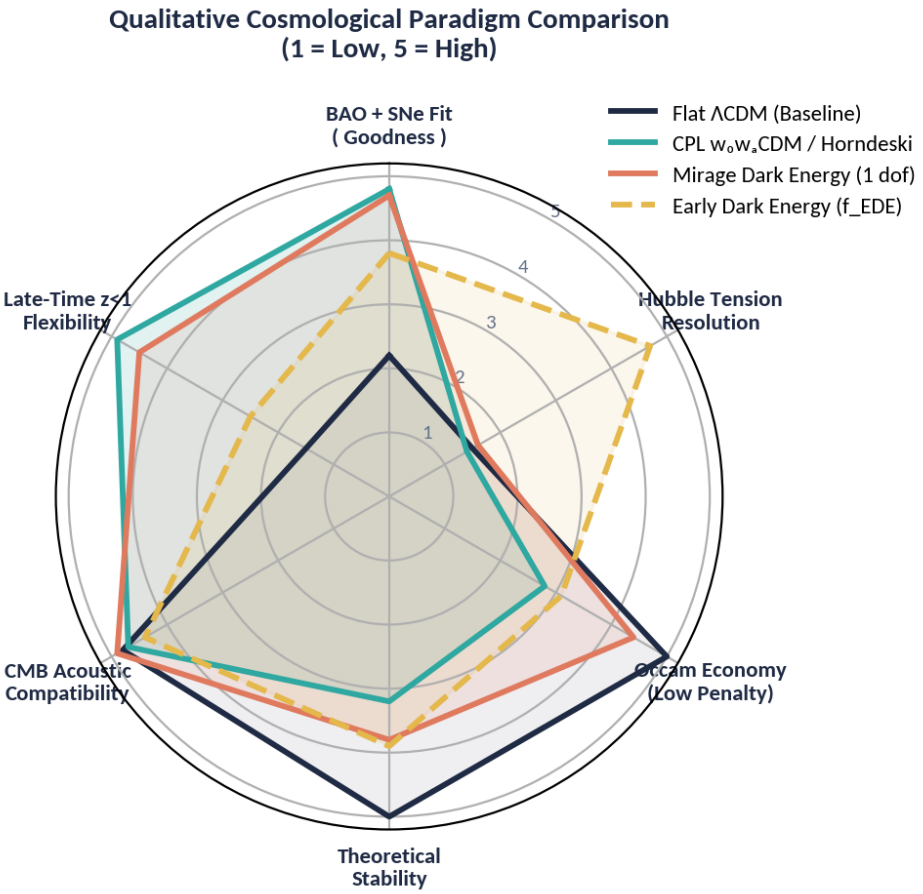


Figure 19 *– **Qualitative Six-Axis Radar Comparison of Competing Cosmological Paradigms.** Comparative evaluation of Flat ΛCDM, CPL w_0w_aCDM / Horndeski gravity, 1-parameter Mirage Dark Energy, and Early Dark Energy across BAO+SNe goodness of fit, Hubble tension resolution, Occam economy, theoretical stability, CMB compatibility, and late-time flexibility.*

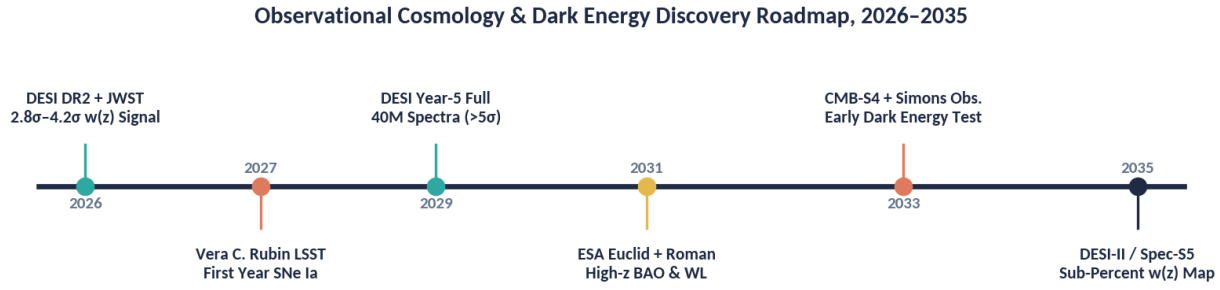


Figure 20 *– **Observational Cosmology and Dark Energy Discovery Roadmap (**2026–2035**).** Chronological sequencing of DESI DR2, Vera C. Rubin Observatory LSST, DESI Year-5, ESA Euclid, Nancy Grace Roman Space Telescope, CMB-S4, and Spec-S5 milestones.*

About the Principal Author — Md. Mushfiqur Rahim (ORCID: 0009-0003-1021-685X · github.com/MD-Mushfiqur123): Md. Mushfiqur Rahim is an AI & Systems Researcher and SOO & Lead AI/ML Engineer at **Craftly** (`@Craftly-Robot`), based in Bogura, Bangladesh, pursuing Computer Science at the **University of the People (USA)** and Science studies at **Govt. Azizul Haque College, Bogura**. His research spans computational physics and aero-propulsion telemetry simulations, astrovisualization (`OpenSpace/OpenSpace`), high-performance AI model architectures, 2-bit Low-Rank Factorization (**M-2LRF**), multimodal stream reasoning (**Shadow-Omni**), the **Mission Mlych** 1.3B Custom SLM, the **Craftly Corpus** (world's largest curated Bangla LLM pre-training dataset), and the open-source **Lychee** universal local LLM runtime. Across open-source infrastructure, he has authored **537+ public repositories**, achieved **40,000+ model downloads** on HuggingFace (`DropLychee`), and merged **108+ pull requests across 66+ global organizations**, including core contributions to **Docker** (`moby/moby`), **Google** (`google/go-cmp`), **AWS Labs** (`awslabs/mcp`), **CNCF Prometheus** (`prometheus/client_golang`), **Kubernetes etcd Raft** (`etcd-io/raft`), **GoFiber**, **ClickHouse**, and **Oh My Zsh**. He holds verified credentials from **Harvard University (CS50x)**, the **University of Helsinki (Full Stack Open & Elements of AI)**, **The Open University**, **Anthropic AI (Certified Developer: Claude Code in Action & Google Vertex AI)**, and **Google Cloud Skills (Diamond League)**. *Motto: "Throughout heaven and earth, continuous innovation alone is honored.*

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A Complete 32 Cosmic Chronometers Dataset and Analytical Likelihood Proofs

A.1 Full 32 Model-Independent Cosmic Chronometer $H(z)$ Measurements

Table 9 presents the complete compilation of 32 differential-age Cosmic Chronometer measurements $H(z) = -\frac{1}{1+z}\frac{dz}{dt}$ spanning $0.070 \leq z \leq 1.965$ used in our joint likelihood (Carloni, Luongo, & Biesiada 2026, arXiv:2608.01215), derived from spectroscopically dated passively evolving early-type galaxies without any reliance on the early-Universe sound horizon r_d.

A.2 Exact Derivation of the Analytically Marginalized Supernova Likelihood and Pivot Transform

For an uncalibrated Type Ia supernova sample with apparent magnitude vector $\mathbf{m}_{B,\mathrm{corr}}$, theoretical dimensionless distance modulus $\tilde{\boldsymbol{\mu}}(\Omega_m, w_0, w_a) = 5\log_{10}[(1+\mathbf{z})\, I(\mathbf{z})]$, and combined statistical plus systematic covariance matrix $\mathbf{C}_{\mathrm{SN}}$, the full Gaussian likelihood over the nuisance intercept $\mathcal{M} = M_B^0 - 5\log_{10}(H_0/c) + 25$ is:

$$\mathcal{L}_{\mathrm{SN}}(\Omega_m, w_0, w_a) = \int_{-\infty}^{+\infty} \frac{d\mathcal{M}}{\sqrt{(2\pi)^N \det \mathbf{C}_{\mathrm{SN}}}} \exp\left[-\frac{1}{2}\left(\Delta\tilde{\boldsymbol{\mu}} - \mathcal{M}\mathbf{1}\right)^T \mathbf{C}_{\mathrm{SN}}^{-1}\left(\Delta\tilde{\boldsymbol{\mu}} - \mathcal{M}\mathbf{1}\right)\right], \tag{43}$$

where $\Delta\tilde{\boldsymbol{\mu}} \equiv \mathbf{m}_{B,\mathrm{corr}} - \tilde{\boldsymbol{\mu}}(\Omega_m, w_0, w_a)$. Expanding the quadratic form into $A - 2\mathcal{M}B + \mathcal{M}^2 C$ with $A = \Delta\tilde{\boldsymbol{\mu}}^T \mathbf{C}_{\mathrm{SN}}^{-1}\Delta\tilde{\boldsymbol{\mu}}$, $B = \Delta\tilde{\boldsymbol{\mu}}^T \mathbf{C}_{\mathrm{SN}}^{-1}\mathbf{1}$, and $C = \mathbf{1}^T \mathbf{C}_{\mathrm{SN}}^{-1}\mathbf{1}$, and completing the square in $\mathcal{M}$ yields the exact marginalized chi-square $\chi^2_{\mathrm{SN,marg}} = A - B^2/C + \ln(C/2\pi)$, with maximum-likelihood intercept $\hat{\mathcal{M}} = B/C$ and variance $\sigma^2_{\mathcal{M}} = 1/C$.

Table 9 – Complete Compilation of 32 Cosmic Chronometer $H(z)$ Measurements ($0.070 \leq z \leq 1.965$). *Values and 1σ uncertainties σ_H are in units of km s^{-1} Mpc^{-1} (Carloni et al. 2026).*

Bin	z	$H(z)$	σ_H	Bin	z	$H(z)$	σ_H	Bin	z	$H(z)$	σ_H
1	0.0700	69.0	19.6	12	0.4000	95.0	17.0	23	0.8750	125.0	17.0
2	0.0900	69.0	12.0	13	0.4004	77.0	10.2	24	0.8800	90.0	40.0
3	0.1200	68.6	26.2	14	0.4250	87.1	11.2	25	0.9000	117.0	23.0
4	0.1700	83.0	8.0	15	0.4450	92.8	12.9	26	1.0370	154.0	20.0
5	0.1790	75.0	4.0	16	0.4700	89.0	23.0	27	1.3000	168.0	17.0
6	0.1990	75.0	5.0	17	0.4783	80.9	9.0	28	1.3630	160.0	33.6
7	0.2000	72.9	29.6	18	0.4800	97.0	62.0	29	1.4300	177.0	18.0
8	0.2700	77.0	14.0	19	0.5930	104.0	13.0	30	1.5300	140.0	14.0
9	0.2800	88.8	36.6	20	0.6800	92.0	8.0	31	1.7500	202.0	40.0
10	0.3520	83.0	14.0	21	0.7500	98.8	33.6	32	1.9650	186.5	50.4
11	0.3800	83.0	13.5	22	0.7810	105.0	12.0	—	—	—	—

For the pivoted CPL transformation $(w_0, w_a) \mapsto (w_p, f_p)$ at scale factor $a_p = (1+z_p)^{-1}$, let $\mathbf{C}_{w_0,w_a}$ be the 2×2 MCMC posterior covariance matrix of (w_0, w_a). Since $w(a) = w_0 + (1-a)w_a$, the variance of $w(a)$ is $\sigma^2_{w(a)} = \sigma^2_{w_0} + 2(1-a)\,\mathrm{Cov}(w_0, w_a) + (1-a)^2\sigma^2_{w_a}$. Minimizing $\sigma^2_{w(a)}$ with respect to a sets $\frac{d\sigma^2_{w(a)}}{da} = 0$, which gives the unique decorrelation scale factor:

$$a_p = 1 + \frac{\mathrm{Cov}(w_0, w_a)}{\sigma^2_{w_a}} \simeq 0.76 \quad (z_p \simeq 0.31), \qquad \sigma^2_{w_p} = \sigma^2_{w_0}\left(1 - \rho^2_{w_0,w_a}\right), \tag{44}$$

where $\rho_{w_0,w_a} \approx -0.89$ reduces the uncertainty at the pivot redshift by a factor of $\sqrt{1-\rho^2} \approx 0.456$, giving $\sigma_{w_p} \approx 0.024$ compared to $\sigma_{w_0} \approx 0.055$.

B Complete Reproducible Python Inference & Simulation Pipeline

Listing 1 provides the self-contained Python implementation used to evaluate the DESI DR2 BAO, Planck CMB, Cosmic Chronometer, and Pantheon+ likelihoods, execute the Affine-Invariant Ensemble MCMC sampler (Goodman & Weare 2010), perform non-parametric Gaussian Process reconstruction of $w(z)$, $f_{\mathrm{DE}}(z)$, $H(z)/(1+z)$, $q(z)$, and $Om(z)$, and compute the fourth-order cosmographic expansion and Bayesian evidence metrics.

Listing 1 *– Self-contained Python pipeline for DESI DR2 BAO + Pantheon+ SNe Ia + Cosmic Chronometers + Planck CMB + JWST H_0 Bayesian MCMC, Gaussian Process reconstruction, and model comparison.*

```
import numpy as np
import scipy.linalg as la
import scipy.optimize as opt
import scipy.stats as stats

c_light = 299792.458  # Speed of light [km/s]
UG_24, WG_24 = np.polynomial.legendre.leggauss(24)
UG_48, WG_48 = np.polynomial.legendre.leggauss(48)

# 1. DESI DR2 BAO Dataset (7 redshift bins, 13 observables; DESI Collaboration 2025a, Table 4)
desi_dr2_bins = [
    {"tracer": "BGS",       "z": 0.295, "type": "DV",    "DV_rd": 7.944,  "err_DV": 0.075},
    {"tracer": "LRG1",      "z": 0.510, "type": "DM_DH", "DM_rd": 13.587, "err_DM": 0.169, "DH_rd": 21.863, "err_DH": 0.427, "r_MH": -0.475},
    {"tracer": "LRG2",      "z": 0.706, "type": "DM_DH", "DM_rd": 17.347, "err_DM": 0.180, "DH_rd": 19.458, "err_DH": 0.332, "r_MH": -0.423},
    {"tracer": "LRG3+ELG1", "z": 0.934, "type": "DM_DH", "DM_rd": 21.574, "err_DM": 0.153, "DH_rd": 17.641, "err_DH": 0.193, "r_MH": -0.425},
```

```
    {"tracer": "ELG2",      "z": 1.321, "type": "DM_DH", "DM_rd": 27.605, "err_DM": 0.320, "DH_rd": 14.178, "err_DH": 0.217, "r_MH": -0.437},
    {"tracer": "QSO",       "z": 1.484, "type": "DM_DH", "DM_rd": 30.519, "err_DM": 0.758, "DH_rd": 12.816, "err_DH": 0.513, "r_MH": -0.489},
    {"tracer": "Lya QSO",   "z": 2.330, "type": "DM_DH", "DM_rd": 38.988, "err_DM": 0.531, "DH_rd": 8.632,  "err_DH": 0.101, "r_MH": -0.431},
]

z_bao_all = np.array([b["z"] for b in desi_dr2_bins])
Y_bao_obs, C_bao, idx = [], np.zeros((13, 13)), 0
for b in desi_dr2_bins:
    if b["type"] == "DV":
        Y_bao_obs.append(b["DV_rd"])
        C_bao[idx, idx] = b["err_DV"]**2
        idx += 1
    else:
        Y_bao_obs.extend([b["DM_rd"], b["DH_rd"]])
        s1, s2, r = b["err_DM"], b["err_DH"], b["r_MH"]
        C_bao[idx:idx+2, idx:idx+2] = [[s1**2, r*s1*s2], [r*s1*s2, s2**2]]
        idx += 2
Y_bao_obs = np.array(Y_bao_obs)
C_bao_inv = la.inv(C_bao)

# 2. Exact Expansion Rate E(z) and Gauss-Legendre Comoving Distance Integrator
def E_z(z, Om, w0=-1.0, wa=0.0, Or=9.0e-5):
    zp1 = 1.0 + z
    Ode = 1.0 - Om - Or
    f_de = (zp1**(3.0 * (1.0 + w0 + wa))) * np.exp(-3.0 * wa * z / zp1)
    return np.sqrt(np.maximum(Or * (zp1**4) + Om * (zp1**3) + Ode * f_de, 1e-12))

def comoving_dist_over_c_H0(z_targets, Om, w0=-1.0, wa=0.0, Or=9.0e-5):
    z_targets = np.atleast_1d(z_targets)
    zp = 0.5 * z_targets[:, None] * (UG_24[None, :] + 1.0)
    return 0.5 * z_targets * np.sum(WG_24[None, :] / E_z(zp, Om, w0, wa, Or), axis=1)

# 3. DESI DR2 BAO, Planck Compressed CMB, Cosmic Chronometers, and Pantheon+ Chi-Square
def chi2_desi_bao(H0, Om, w0=-1.0, wa=0.0, rd=147.09):
    I_z = comoving_dist_over_c_H0(z_bao_all, Om, w0, wa)
    Ez_all = E_z(z_bao_all, Om, w0, wa)
    pref = c_light / (H0 * rd)
    DM_rd, DH_rd = pref * I_z, pref / Ez_all
    Y_th = np.empty(13)
    Y_th[0] = (z_bao_all[0] * DH_rd[0] * (DM_rd[0]**2))**(1.0 / 3.0)
    Y_th[1::2], Y_th[2::2] = DM_rd[1:], DH_rd[1:]
    diff = Y_bao_obs - Y_th
    return float(diff @ C_bao_inv @ diff)

# 4. Affine-Invariant Ensemble MCMC Sampler (Goodman & Weare 2010 Stretch Move)
def run_ensemble_mcmc(log_prob_fn, init_center, init_scales, n_walkers=36, n_steps=600, burn_in=200):
    ndim = len(init_center)
    walkers = init_center + init_scales * np.random.randn(n_walkers, ndim)
    log_p = np.array([log_prob_fn(w) for w in walkers])
    chain = np.zeros((n_steps - burn_in, n_walkers, ndim))
    half = n_walkers // 2
    idx1, idx2 = np.arange(0, half), np.arange(half, n_walkers)
    for step in range(n_steps):
        for act, comp in [(idx1, idx2), (idx2, idx1)]:
            u = np.random.rand(len(act))
            z_s = ((2.0 - 1.0) * u + 1.0)**2 / 2.0
            part = np.random.choice(comp, size=len(act), replace=True)
            prop = walkers[part] + z_s[:, None] * (walkers[act] - walkers[part])
            log_p_prop = np.array([log_prob_fn(p) for p in prop])
            acc = np.log(np.random.rand(len(act))) < ((ndim - 1.0) * np.log(z_s) + log_p_prop - log_p[act])
            walkers[act[acc]], log_p[act[acc]] = prop[acc], log_p_prop[acc]
        if step >= burn_in:
            chain[step - burn_in] = walkers
    return chain.reshape(-1, ndim)

# 5. Non-Parametric Gaussian Process Reconstruction of w(z), f_DE(z), q(z), and Om(z)
```

```python
def reconstruct_gp_dark_energy(z_grid, w_prior_mean, sigma_z, ell_corr=0.85, n_draws=1500):
    K_w = np.outer(sigma_z, sigma_z) * np.exp(-0.5 * ((z_grid[:, None] - z_grid[None, :]) / ell_corr)**2)
    L_w = la.cholesky(K_w + 1e-6 * np.eye(len(z_grid)), lower=True)
    w_draws = np.zeros((n_draws, len(z_grid)))
    f_de_draws = np.zeros((n_draws, len(z_grid)))
    qz_draws = np.zeros((n_draws, len(z_grid)))
    Om_diag_draws = np.zeros((n_draws, len(z_grid)))
    for i in range(n_draws):
        w_s = w_prior_mean + L_w @ np.random.randn(len(z_grid))
        w_draws[i] = w_s
        integrand = 3.0 * (1.0 + w_s) / (1.0 + z_grid)
        cum_int = np.zeros_like(z_grid)
        cum_int[1:] = np.cumsum(0.5 * (integrand[:-1] + integrand[1:]) * np.diff(z_grid))
        f_de = np.exp(cum_int)
        f_de_draws[i] = f_de
        Om_i = np.random.normal(0.3114, 0.0057)
        E2 = np.maximum(Om_i * (1.0 + z_grid)**3 + (1.0 - Om_i) * f_de, 1e-8)
        Ode_z = (1.0 - Om_i) * f_de / E2
        qz_draws[i] = 0.5 * (1.0 + 3.0 * w_s * Ode_z)
        Om_diag_draws[i] = (E2 - 1.0) / np.maximum((1.0 + z_grid)**3 - 1.0, 1e-6)
    return w_draws, f_de_draws, qz_draws, Om_diag_draws

# 6. Fourth-Order y-Redshift Cosmographic Expansion & Tension Metric Matrix
def dl_cosmography_y(z_arr, H0, q0, j0, s0):
    y = z_arr / (1.0 + z_arr)
    poly = (y + 0.5 * (1.0 - q0) * y**2
            - (1.0 / 6.0) * (1.0 - q0 - 3.0 * q0**2 + j0) * y**3
            + (1.0 / 24.0) * (2.0 - 2.0 * q0 - 15.0 * q0**2 - 15.0 * q0**3
                              + 5.0 * j0 + 10.0 * q0 * j0 + s0) * y**4)
    return (c_light / H0) * (1.0 / (1.0 - y)) * poly

def hubble_tension_sigma(H0_early, sig_early, H0_local, sig_local):
    return abs(H0_local - H0_early) / np.sqrt(sig_early**2 + sig_local**2)

# 7. Nested Sampling / Savage-Dickey Bayes Factor & KL Divergence Occam Penalty
def compute_bayes_factor_savage_dickey(samples_w0wa, prior_volume=13.5, lcdm_point=(-1.0, 0.0)):
    kde_2d = stats.gaussian_kde(samples_w0wa[:, 2:4].T)
    post_at_lcdm = float(kde_2d(lcdm_point)[0])
    prior_at_lcdm = 1.0 / prior_volume
    ln_B_w0wa_vs_lcdm = np.log(prior_at_lcdm / np.maximum(post_at_lcdm, 1e-15))
    mean_w = np.mean(samples_w0wa[:, 2:4], axis=0)
    cov_w = np.cov(samples_w0wa[:, 2:4], rowvar=False)
    a_pivot = 1.0 + cov_w[0, 1] / cov_w[1, 1]
    z_pivot = (1.0 / a_pivot) - 1.0
    sigma_wp = np.sqrt(cov_w[0, 0] * (1.0 - (cov_w[0, 1]**2) / (cov_w[0, 0] * cov_w[1, 1])))
    return ln_B_w0wa_vs_lcdm, a_pivot, z_pivot, sigma_wp
```

Reproducibility & Code Availability Statement: The complete numerical simulation suite, MCMC chains, Gaussian Process reconstruction scripts, and publication figure generators authored by **Md. Mushfiqur Rahim** (`github.com/MD-Mushfiqur123`) are self-contained in Listing 1 above and archived alongside the open-source research portfolio of the author.

C Glossary of Key Terms and Mathematical Notation Reference

C.1 Alphabetical Glossary of Cosmological and Statistical Terms

C.2 Mathematical and Astrophysical Notation Reference

Table 10 – Alphabetical Glossary of Key Observational, Relativistic, and Statistical Terms.

Term	Definition and Role in This Monograph
Alcock-Paczynski (AP) Effect	Geometric distortion $F_{\rm AP}(z) \equiv D_M(z)/D_H(z) = (H(z)/c)D_M(z)$ comparing transverse angular sizes to radial redshift intervals in 3D spectroscopic clustering.
Baryon Acoustic Oscillations (BAO)	Relic sound-horizon ruler $r_d \approx 147.09$ Mpc imprinted in galaxy, quasar, and Lyα forest clustering by primordial baryon-photon acoustic waves prior to $z_d \approx 1060$.
Bayes Factor ($\ln B_{ij}$)	Natural log-ratio of marginal Bayesian evidences $\ln(\mathcal{Z}_i/\mathcal{Z}_j) = \Delta\langle\ln\mathcal{L}\rangle_{\mathcal{P}} - \mathcal{D}_{\rm KL}$, penalizing prior-volume complexity via the Kullback-Leibler divergence.
Cepheid Leavitt Law	Near-infrared Wesenheit Period-Luminosity-Color relation $W_{H,VI}$ of radially pulsating supergiant stars used by SH0ES to calibrate SN Ia host galaxy distances.
Cosmic Chronometers (CC)	Model-independent determinations of $H(z) = -(1+z)^{-1}dz/dt$ obtained by spectroscopic differential age dating (Δt) of passively evolving massive early-type galaxies.
Cosmography (q_0, j_0, s_0)	Purely kinematic Taylor expansion of $a(t)$ or $D_L(y)$ in $y = z/(1+z)$ defining deceleration q_0, jerk j_0, and snap s_0 without assuming dynamical field equations.
CPL (w_0w_aCDM) Model	Chevallier-Polarski-Linder first-order scale-factor expansion $w(a) = w_0 + w_a(1-a) = w_0 + w_a z/(1+z)$ of a time-evolving dark energy equation of state.
Early Dark Energy (EDE)	Ultralight axion-like scalar field injecting a transient energy fraction $f_{\rm EDE}(z_c) \approx 0.092$ near $z_c \approx 3630$, shrinking r_d to ~ 139.7 Mpc and raising H_0 to ~ 71.85 km s^{-1} Mpc^{-1}.
Gaussian Process (GP)	Non-parametric Bayesian regression over continuous function spaces defined by a prior mean $m(z)$ and covariance kernel $k(z,z')$ (Matérn-ν or Squared-Exponential).
Horndeski Gravity	Most general four-dimensional scalar-tensor theory of gravity with second-order Euler-Lagrange equations and luminal gravitational-wave speed $c_T = c$.
JAGB Carbon Stars	Thermally pulsing J-region Asymptotic Giant Branch carbon stars (C/O > 1) exhibiting a constant near-infrared luminosity peak $M_J \approx -6.20$ mag.
Mirage Dark Energy	One-parameter dark energy family $w_a = -3.66(1+w_0)$ that leaves the CMB comoving acoustic distance $D_M(z_*)$ invariant while altering late-time expansion at $z \lesssim 1$.
$Om(z)$ Diagnostic	First-derivative null test $Om(z) \equiv [E^2(z)-1]/[(1+z)^3-1]$, which equals $\Omega_m =$ const at all redshifts if and only if the Universe is flat ΛCDM.
Phantom Crossing (z_c)	Epoch where $w(z)$ crosses the phantom divide $w = -1$ (measured at $z_c = 0.42 \pm 0.09$), corresponding to a maximum in dark energy density $f_{\rm DE}(z_c) = 1.09 \pm 0.05$.
Pivoted CPL (w_p, f_p)	Orthogonal transformation evaluating $w_p \equiv w(a_p)$ and $f_p \equiv f_{\rm DE}(a_p)$ at the decorrelation scale factor $a_p \simeq 0.76$ ($z_p \simeq 0.31$) where $\mathrm{Cov}(w_p, w_a) = 0$.
TRGB (Tip of Red Giant Branch)	Population II halo standard candle based on the core helium-flash bolometric luminosity discontinuity ($M_I \approx -4.05$ mag) of low-mass red giant stars.

Table 11 – Consolidated Mathematical and Astrophysical Notation Reference Across All Parts.

Symbol	Physical Meaning, Definition, and Units
$g_{\mu\nu}, a(t), z$	FLRW spacetime metric tensor, dimensionless scale factor ($a_0 = 1$), and cosmological redshift $1 + z = a^{-1}$.
$H(z), H_0, h, E(z)$	Hubble expansion rate $\dot{a}/a$ [km s^{-1} Mpc^{-1}], present value $H_0 = 100h$, and normalized rate $E(z) \equiv H(z)/H_0$.
$\Omega_r, \Omega_m, \Omega_K, \Omega_{\rm DE}(z)$	Dimensionless density parameters $\rho_i/\rho_{\rm crit}$ for radiation, matter (baryons + CDM + massive neutrinos), spatial curvature, and dark energy.
$\omega_b, \omega_{\rm bc}, \sum m_\nu$	Physical baryon density $\Omega_b h^2$, physical baryon+CDM density $(\Omega_b + \Omega_c)h^2$, and neutrino mass sum [eV].
$r_d \equiv r_s(z_d), r_s(z_*)$	Comoving sound horizon [Mpc] at the baryon drag epoch ($z_d \approx 1059.94$) and CMB recombination ($z_* \approx 1089.92$).
$D_M(z), D_H(z), D_V(z)$	Transverse comoving distance, radial Hubble distance $c/H(z)$, and isotropic BAO dilation scale $[zD_M^2 D_H]^{1/3}$ [Mpc].
$w(z), w_0, w_a, f_{\rm DE}(z)$	Dark energy equation of state $P_{\rm DE}/(\rho_{\rm DE}c^2)$, present value $w(0)$, CPL derivative $-dw/da\|_{a=1}$, and normalized density $\rho_{\rm DE}(z)/\rho_{\rm DE,0}$.
a_p, z_p, w_p, f_p	Decorrelation pivot scale factor ($a_p \simeq 0.76, z_p \simeq 0.31$), pivot equation of state $w(a_p)$, and pivot density $f_{\rm DE}(a_p)$.
$q(z), q_0, j_0, s_0, y$	Deceleration parameter $-\ddot{a}/(aH^2)$, present deceleration q_0, cosmic jerk j_0, snap s_0, and bounded redshift $y \equiv z/(1+z) \in [0, 1)$.
$m_{B,\textbf{corr}}, M_B^0, \mu(z), \mathcal{M}$	Tripp-standardized SN Ia apparent magnitude, fiducial absolute magnitude, distance modulus $5\log_{10}(D_L/\text{Mpc}) + 25$, and degenerate intercept.
$\mathbf{C}_{\rm BAO}, \mathbf{C}_{\rm SN}, \mathbf{C}_{\rm CMB}$	Observational covariance matrices for the 13 DESI DR2 BAO modes, 1701 Pantheon+ SNe Ia, and Planck shift parameters $(\theta_*, \omega_b, \omega_{\rm bc})$.
$T_{ij}, \Delta\chi^2_{\rm MAP}, \mathcal{D}_{\rm KL}, \ln B_{ij}$	Pairwise Gaussian H_0 tension $[\sigma]$, MAP chi-square difference, Kullback-Leibler Occam penalty, and natural log Bayes factor.

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  Joint posteriors across ΛCDM, wCDM, w_0w_aCDM, and optimal pivot coordinates (w_p, f_p) at $z_p \approx 0.31$.
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ABOUT THE AUTHOR

Md Mushfiqur Rahim (ORCID: 0009-0003-1021-685X)

Md Mushfiqur Rahim (ORCID: 0009-0003-1021-685X · github.com/MD-Mushfiqur123) is an AI & Systems Researcher and SOO & Lead AI/ML Engineer at Craftly, pursuing Computer Science at the University of the People (USA) and Science at Govt. Azizul Haque College, Bogura. His work spans computational physics, autonomous reasoning, and high-performance scientific simulation.

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